Analysis of brain network models for Alzheimer's disease via spreading of prion-like misfolded tau proteins

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Abstract

In this article we study the onset and progression of Alzheimer's disease via three models of increasing complexity and all based on the prion-like hypothesis: Fisher-Kolmogorov (FK), heterodimer (HD) and Smoluchowski (SMO) models. These are initially defined in a continuous setting, and later discretized both in space, with graph theory, and time, with standard Euler methods; the former also requires a study on timescales to balance the different phenomena at play. Afterwards, an asymptotic analysis is discussed for plot normalization and a function to estimate the convergence time is empirically proposed. Finally, all models are complicated by considering time-varying diffusion, whereas for the SMO one the accumulation and size dependency are studied as well; all results found are in line with our expectations based on the theory developed, although some require additional theory to be fully understood.

1 Introduction

Alzheimer's disease (AD) is one of the most ill-famed neurodegenerative disease, whose devastating condition is associated with cognitive decline and eventually death.

Post-mortem analyses of brain tissue reveal the presence of extracellular amyloid-beta ($A\beta$) plaques and intracellular neurofibrillary tangles of tau (τ) proteins; indeed, these aggregates have been observed and correlated with the evolution of the disease following a predictable spreading pattern through the brain. However, even though these two proteins are known to interact with each other, [15, § 3 on p. 3] in this paper we only focus on the role of τ proteins in AD.

These systematic invasion patterns of protein aggregates are the basis of the prion-like hypothesis for AD:

- H1 AD is caused by misfolded proteins;
- H2 misfolded proteins act as a toxic template on which regular τ proteins can be bound and converted to misfolded ones;
- H3 misfolded proteins can form aggregates of increasing size, which may also fragment into smaller aggregates;
- H4 misfolded proteins and its aggregates spread primarily across the brain through the network of axonal pathways.

These mechanisms are based on prion diseases firstly observed in Bovine Spongiform Encephalopathy (BSE), more commonly known as mad cow disease, and can be applied more broadly to other neurodegenerative diseases (for instance Parkinson's).

Bear in mind this article focuses only on the dynamic of misfolded τ proteins and not on their damage to the brain, which would require another set of equations to predict brain atrophy using the toxic protein concentrations (see [3] as an example).

2 Kinetic models

To predict the concentration of misfolded τ proteins, Three continuous models are considered, followed by spatial and temporal discretization methods. Afterwards, an asymptotic analysis is discussed for plot normalization and complications are introduced to the base models.

We assume the brain is contained in the closed region $\mathcal{B} \subset \mathbb{R}^3$ and that it's observed in a time interval $\mathcal{T} \equiv (0,T]$ of duration $T \in \mathbb{R}^+_*$, where we set

$$\begin{aligned} \mathbb{R}^+ &\equiv \{x \in \mathbb{R} : x \geq 0\}, \\ \mathbb{R}^+_* &\equiv \{x \in \mathbb{R} : x > 0\}, \end{aligned} \tag{1}$$

as the positive real numbers with and without the origin. It's also handy to define

$$\begin{aligned} \mathcal{Q} &\equiv \mathcal{B} \times \mathcal{T}, && \text{(space-time cylinder)} \\ \partial\mathcal{Q}_x &\equiv \partial\mathcal{B} \times \mathcal{T}, && \text{(space boundary)} \\ \partial\mathcal{Q}_t &\equiv \mathcal{B} \times \partial\mathcal{T}_-, && \text{(time boundary)} \end{aligned} \tag{2}$$

with $\partial\mathcal{T}_- \equiv \{0\}$.

2.1 Continuous models

These models exhibit different levels of complexity depending on which assumptions are adopted and how they are implemented: all three considers H1 and H2 but only the last one takes into account H3. [15, 16]

However, a shared and fundamental feature is the non-linear nature due to the quadratic reaction term necessary to model H2.

2.1.1 The heterodimer model (HD)

As the name suggests, it considers only two possible protein configurations: a natural healthy state with concentration $c_h \in \mathcal{C}^2_+(\mathcal{Q})$ and a misfolded one with $c_m \in \mathcal{C}^2_+(\mathcal{Q})$, where

$$\mathcal{C}^2_+(\mathcal{Q}) \equiv \left\{ f \in \mathcal{C}^2(\mathcal{B}) \cap \mathcal{C}^1(\mathcal{T}) \mid f \geq 0 \text{ in } \mathcal{Q} \right\}.$$

Thus, the model is described by a system of two ordinary differential equations (ODEs):

$$[\mathrm{HD}]_c \begin{cases} \left.\begin{aligned} \frac{\mathrm{d}c_h}{\mathrm{d}t} &= \Delta^{\mathbf{D}} c_h + \gamma - \mu_h c_h - \kappa c_h c_m \\ \frac{\mathrm{d}c_m}{\mathrm{d}t} &= \Delta^{\mathbf{D}} c_m \qquad - \mu_m c_m + \kappa c_h c_m \end{aligned}\right\} \text{ in } \mathcal{Q}, \\ \text{BC: } c_h = c_m = 0, & \text{on } \partial\mathcal{Q}_x, \\ \text{IC: } c_h = c_{h,0},\ c_m = c_{m,0} \neq 0 & \text{on } \partial\mathcal{Q}_t, \end{cases}$$

where $\Delta^{\mathbf{D}} f \equiv \nabla \cdot (\mathbf{D} \cdot \nabla f)$ is the tensorial Laplacian operator[1], with the diffusion tensor $\mathbf{D}$, $\gamma \in \mathbb{R}^+$ is the production rate of healthy protein, $\mu_h \in \mathbb{R}^+$ and $\mu_m \in \mathbb{R}^+$ are the clearance rates of healthy and misfolded proteins respectively, and $\kappa \in \mathbb{R}^+$ is the misfolding conversion rate.

2.1.2 The Fisher-Kolmogorov model (FK)

It can be derived from $[\mathrm{HD}]_c$ by assuming c_h is homogeneous and stationary, and remains so throughout the dynamic (see Appendix A.1 for more details).

It's the simplest one since it's based only on the misfolded protein concentration $c \in \mathcal{C}^2_+(\mathcal{Q})$ with one nonlinear reaction-diffusion ODE:

$$[\mathrm{FK}]_c \begin{cases} \dfrac{\mathrm{d}c}{\mathrm{d}t} = \Delta^{\mathbf{D}} c + \alpha c(1-c) \text{ in } \mathcal{Q}, \\ \text{BC: } c = 0 & \text{on } \partial\mathcal{Q}_x, \\ \text{IC: } c = c_0 \neq 0 & \text{on } \partial\mathcal{Q}_t, \end{cases}$$

where c is a dimensionless concentration and $\alpha \geq 0$ represents the misfolding conversion rate.

2.1.3 The Smoluchowski model (SMO)

It's the most nuanced model as it consider toxic chains: set $c_1 \in \mathcal{C}^2_+(\mathcal{Q})$ as the concentration of healthy monomers and $c_i \in \mathcal{C}^2_+(\mathcal{Q})$ as the one of misfolded polymers of size $1 < i \in \{1, 2, \dots\} \equiv \mathbb{N}^+$, then H3 can be expressed with the reaction equation

$$c_i + c_j \underset{f_{ij}}{\overset{a_{ij}}{\rightleftharpoons}} c_{ij},$$

where $a_{ij} \in \mathbb{R}^+$ and $f_{ij} \in \mathbb{R}^+$ are respectively the aggregation and fragmentation rates. Hence, the collective effect of aggregation and fragmentation can be summarized by

$$\begin{aligned} A_i &= A_i^g - A_i^l = \frac{1}{2} \sum_{j=1}^{i-1} a_{j(i-j)} c_j c_{i-j} - \sum_{j=1}^{\infty} a_{ij} c_i c_j, \\ F_i &= F_i^l - F_i^g = \frac{1}{2} \sum_{j=1}^{i-1} f_{j(i-j)} c_i - \sum_{j=1}^{\infty} f_{ij} c_{i+j}, \end{aligned} \tag{5}$$

in which A_i^g and F_i^g denote the gain terms for the i-th polymer c_i due to aggregation and fragmentation, while A_i^l and F_i^l are the corresponding loss terms.

The full Smoluchowski model is thus a non-linear system of *infinite* reaction-diffusion ODEs:

$$\begin{cases} \dfrac{\mathrm{d}c_i}{\mathrm{d}t} = \Delta^{\mathbf{D}} c_i + \gamma_i - \mu_i c_i + A_i - F_i \ \forall i \in \mathbb{N}^+, \text{ in } \mathcal{Q}, \\ \text{BC: } c_i = 0 \ \forall i \in \mathbb{N}^+ & \text{on } \partial\mathcal{Q}_x, \\ \text{IC: } c_1(0) = c_{1,0},\ c_i(0) = 0 \ \forall i \in \mathbb{N}^+ \setminus \{1\}, & \text{on } \partial\mathcal{Q}_t, \end{cases} \tag{6}$$

where $\gamma_i \in \mathbb{R}^+$ and $\mu_i \in \mathbb{R}^+$ have the usual meaning of production and clearance rate already seen in $[\mathrm{HD}]_c$.

Of course, since (6) is in fact intractable, assumptions are needed to simplify the system of ODEs:

- A1 the largest polymer length is $L \in \mathbb{N}^+$ which can neither aggregate nor fragment:
  $$a_{ij} = f_{ij} = 0 \text{ with } i = L \text{ or } j = L;$$
- A2 the diffusion tensor is size-independent, that is
  $$\mathbf{D}_i \equiv \mathbf{D}, \text{ with } 1 \leq i \leq L;$$
- A3 production is only possible with healthy monomers:
  $$\gamma_i = 0 \ \forall i \in \mathbb{N}^+ \setminus \{1\}$$
  while $\gamma_1 = \gamma \in \mathbb{R}^+$;
- A4 clearance is size-independent with rate
  $$\mu_i = \mu \in \mathbb{R}^+, \text{ with } 1 \leq i \leq L;$$
- A5 nucleation takes place at a rate $\kappa \equiv a_{11}$ and is irreversible, namely dimers are stable;
- A6 aggregation is size-independent, that is
  $$a_{ij} = a \in \mathbb{R}^+, \text{ with } 1 \leq i, j < L,$$
  but can only occur by adding single monomers:
  $$a_{ij} = 0, \text{ with } 1 < i, j < L;$$

[1] $\Delta^{\mathbf{D}}$ is equal to the classical Laplacian operator iff $\mathbf{D}$ is isotropic: if $\mathbf{D} = D\mathbf{I}$, with $D \in \mathbb{R}^+$, then $\Delta^{\mathbf{D}} f = D \Delta f$.

A7 fragmentation into monomers is impossible, *i.e.*

$$f_{ij}=0, \text{with } i=1 \text{ or } j=1,$$

whereas fragmentation into smaller polymers is size-independent:

$$f_{ij}=f\in\mathbb{R}^{+}, \text{with } 1<i,j<L.$$

These assumptions lead from (6) to

$$[\mathrm{SMO}]_c\begin{cases}\left.\begin{aligned}&\frac{\mathrm{d}c_1}{\mathrm{d}t}=\Delta^{\mathbf{D}}c_1+\gamma-\mu c_1-2\boldsymbol{\kappa}c_1^2-ac_1\sum_{j=2}^{L-1}c_j\\&\frac{\mathrm{d}c_2}{\mathrm{d}t}=\Delta^{\mathbf{D}}c_2-\mu c_2+\boldsymbol{\kappa}c_1^2-ac_1c_2+f\sum_{j=2}^{L-3}c_{2+j}\\&\frac{\mathrm{d}c_i}{\mathrm{d}t}=\Delta^{\mathbf{D}}c_i-\mu c_i+ac_1c_{i-1}-ac_1c_i\\&\qquad-\frac{(i-3)}{2}fc_i+f\sum_{j=2}^{L-i-1}c_{i+j}, 2<i<L\\&\frac{\mathrm{d}c_L}{\mathrm{d}t}=\Delta^{\mathbf{D}}c_L-\mu c_L+ac_1c_{L-1}\end{aligned}\right\}\text{in } \mathcal{Q},\\ \text{BC: } c_i=0,\ 1\leq i\leq L, \qquad\qquad \text{on } \partial\mathcal{Q}_x,\\ \text{IC: } c_1=c_{1,0},\ c_i=0,\ 1<i\leq L, \qquad \text{on } \partial\mathcal{Q}_t,\end{cases}$$

where the summation $\sum_{j=2}^{L-i-1}$ is valid iff

$$L-i-1\geq 2 \implies i\leq L-3,$$

hence it's not present in the c_{L-2} and c_{L-1} equations; moreover we assume $\boldsymbol{\kappa}\in L_{+}^{\infty}(\mathcal{B})$, with

$$L_{+}^{\infty}(\mathcal{B})\equiv\{f\in L^{\infty}(\mathcal{B})\mid f\geq 0 \text{ in } \mathcal{B}\},$$

is the nucleation rate function[2] assumed here to have form

$$\boldsymbol{\kappa}(\mathbf{x})\equiv\kappa\mathbb{1}_{\mathcal{K}}(\mathbf{x})=\begin{cases}\kappa & \text{if } \mathbf{x}\in\mathcal{K},\\ 0 & \text{otherwise},\end{cases} \tag{8}$$

with $\kappa\in\mathbb{R}^{+}$ and $\mathcal{K}\subseteq\mathcal{B}$.

2.2 Discrete models

For the spatial discretization graph theory is applied to take full advantage of H4; in addition, characteristic times have to be considered for a biologically sound dynamic.

Instead, for the temporal discretization Euler methods, both explicit and implict, are used.

2.2.1 Spatial discretization

One of the most common representation of the brain is with an undirected loopless network, where nodes correspond to neurons and weighted edges to axons; indeed, the same topology is perfect to model its anisotropic diffusion, found in $\Delta^{\mathbf{D}}$.

Given V vertices, two important matricies can be defined:

- the adjacency matrix $\mathbf{A}$ whose entries

  $$A_{ij}\in\mathbb{R}^{+}, \text{with } 1\leq i,j\leq V,$$

  indicate the edge weight connecting the i-th node to the j-th one: if $A_{ij}=0$ it means the two nodes are unconnected;
- the degree matrix

  $$\mathbf{D}\equiv\mathrm{diag}(\mathbf{A}\mathbf{1}),$$

  where $\mathbf{1}$ is the unitary vector in $\mathbb{R}^{V}$: it's a diagonal matrix whose entries

  $$D_{ii}\in\mathbb{R}^{+}, \text{with } 1\leq i\leq V,$$

  are the sum of the rows of $\mathbf{A}$.

The difference of the latter with the former is defined as the Laplacian matrix:

$$\mathbf{L}\equiv\rho(\mathbf{D}-\mathbf{A})=\rho\boldsymbol{\mathcal{L}}, \tag{9}$$

where ρ is a constant to balance the diffusion and growth phenomena, as is going to be explained shortly in § 2.2.2. The Laplacian matrix is the discretization of the tensorial Laplacian operator: $-\mathbf{L}\longleftrightarrow\Delta^{\mathbf{D}}$; indeed, as an example, remember that

$$\Delta^{\mathbf{D}}c=0 \iff c(\mathbf{x},t)=c\ \forall\mathbf{x}\in\mathbb{R}^3,\ t \text{ fixed},$$

and this behaviour is perfectly reproduced with $\mathbf{L}$ if applied to a uniform vector $\mathbf{c}=c\mathbf{1}$:

$$\mathbf{L}\mathbf{c}=c\mathbf{L}\mathbf{1}=c\rho\mathcal{L}\mathbf{1}=c\rho(\mathbf{D}\mathbf{1}-\mathbf{A}\mathbf{1})=c\rho(\mathbf{A}\mathbf{1}-\mathbf{A}\mathbf{1})=\mathbf{0},$$

with $c\in\mathbb{R}^{+}$.

The choice of V is quite critical because it's fundamentally arbitrary as it depends on the level of accuracy necessary and computational power available; for our purposes it's sufficient to consider $V=83$ vertices corresponding to the 83 regions of interest (ROIs) in the brain. In each ROI the average concentration is assumed to be homogeneous in space and its value can thus be represented as a node.

To build the Laplacian one can start from a more fine brain network representation [2, 17, 18]; nonetheless, for the sake of simplicity, in this article the Laplacian used in [6, 15], shown in Fig. 1, is considered.

By collecting all ROI concentrations in the vector

$$\mathbf{c}=[c^1,\dots,c^V]^{\top},$$

[2] In the continuous setting mollification may be necessary to preserve at least continuity of the product κc_1^2; in the weak formulation, however, $L_{+}^{\infty}(\mathcal{B})$ is more than sufficient. In addition, here, the graph-based spatial discretization makes c_1 piecewise continuous, so continuity is not required.

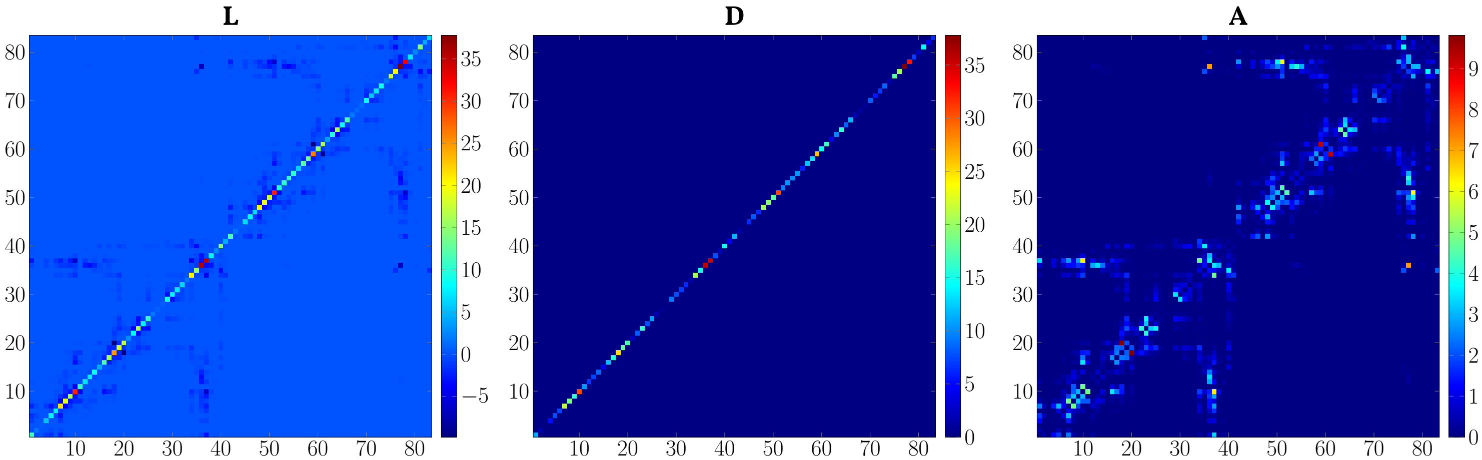

Figure 1: Matrices used the spatial discretization.

whose reconstructed piecewise continuous function

$$\tilde{c}(\mathbf{x},t)=\sum_{i=k}^{V}c^k(t)\mathbb{1}_{\mathrm{ROI}_k}(\mathbf{x})\approx c(\mathbf{x},t) \tag{10}$$

is no longer in $\mathcal{C}^2_+(\mathcal{Q})$ but now lies in $\tilde{c}\in L^\infty_+(\mathcal{B})\cap\mathcal{C}^1(\mathcal{T})$, the three models seen before are discretized as follows: for $[\mathrm{FK}]_c$

$$[\mathrm{FK}]_d\begin{cases}\dfrac{\mathrm{d}\mathbf{c}}{\mathrm{d}t}=-\mathbf{L}\mathbf{c}+\alpha\mathbf{c}\odot(1-\mathbf{c}) & \text{in } \mathcal{T},\\ \text{IC: } \mathbf{c}=\mathbf{c}_0=c_0\mathbf{q}\neq\mathbf{0} & \text{in } \partial\mathcal{T}_-;\end{cases}$$

for $[\mathrm{HD}]_c$

$$[\mathrm{HD}]_d\begin{cases}\left.\begin{aligned}\dfrac{\mathrm{d}\mathbf{c}_h}{\mathrm{d}t}&=-\mathbf{L}\mathbf{c}_h+\gamma\mathbf{1}-\mu_h\mathbf{c}_h-\kappa\mathbf{c}_h\odot\mathbf{c}_m\\ \dfrac{\mathrm{d}\mathbf{c}_m}{\mathrm{d}t}&=-\mathbf{L}\mathbf{c}_m\qquad\quad-\mu_m\mathbf{c}_m+\kappa\mathbf{c}_h\odot\mathbf{c}_m\end{aligned}\right\} & \text{in } \mathcal{T},\\ \text{IC: } \mathbf{c}_h=c_{h,0}\mathbf{1},\ \mathbf{c}_m=c_{m,0}\mathbf{q}\neq\mathbf{0} & \text{in } \partial\mathcal{T}_-;\end{cases}$$

finally for $[\mathrm{SMO}]_c$

$$[\mathrm{SMO}]_d\begin{cases}\left.\begin{aligned}\dfrac{\mathrm{d}\mathbf{c}_1}{\mathrm{d}t}&=-\mathbf{L}\mathbf{c}_1+\gamma\mathbf{1}-\mu\mathbf{c}_1-2\boldsymbol{\kappa}\odot\mathbf{c}_1^2-a\mathbf{c}_1\odot\sum_{j=2}^{L-1}\mathbf{c}_j\\ \dfrac{\mathrm{d}\mathbf{c}_2}{\mathrm{d}t}&=-\mathbf{L}\mathbf{c}_2-\mu\mathbf{c}_2+\boldsymbol{\kappa}\odot\mathbf{c}_1^2-a\mathbf{c}_1\odot\mathbf{c}_2+f\sum_{j=2}^{L-3}\mathbf{c}_{2+j}\\ \dfrac{\mathrm{d}\mathbf{c}_i}{\mathrm{d}t}&=-\mathbf{L}\mathbf{c}_i-\mu\mathbf{c}_i+a\mathbf{c}_1\odot\mathbf{c}_{i-1}-a\mathbf{c}_1\odot\mathbf{c}_i\\ &\quad-\frac{(i-3)}{2}f\mathbf{c}_i+f\sum_{j=2}^{L-i-1}\mathbf{c}_{i+j},\ 2<i<L\\ \dfrac{\mathrm{d}\mathbf{c}_L}{\mathrm{d}t}&=-\mathbf{L}\mathbf{c}_L-\mu\mathbf{c}_L+a\mathbf{c}_1\odot\mathbf{c}_{L-1}.\end{aligned}\right\} & \text{in } \mathcal{T},\\ \text{IC: } \mathbf{c}_1=c_{1,0}\mathbf{1},\ \mathbf{c}_i=\mathbf{0},\ \forall i>2 & \text{in } \partial\mathcal{T}_-,\end{cases}$$

where $\mathbf{c}_i^2\equiv\mathbf{c}_i\odot\mathbf{c}_i$ and $\boldsymbol{\kappa}=\kappa\mathbf{q}$ is the infected ROI vector.

In all previous models the time derivative $\mathrm{d}/\mathrm{d}t$ is applied element-wise exactly like the product $\odot$ (Hadamard product); in addition, $\mathbf{q}\in\{0,1\}^V$ is a vector with q unitary entries describing which nodes have the initial toxic mass or where nucleation takes place.

Observe how the number of unknowns is just the number of concentration values multiplied by the number of vertices: V, $2V$ and LV respectively; indeed, thanks to the discretization in space, each function went from having an infinite amount of degrees of freedom to at most LV.

Lastly, the Dirichlet homogeneous boundary conditions are not present in $[\mathrm{FK}]_d$, $[\mathrm{HD}]_d$ and $[\mathrm{SMO}]_d$ any more, because their value is averaged in the respective ROI.

2.2.2 Characteristic times

In all models previously seen, for misfolded τ proteins there are two important phenomena at play:

- diffusion depending on the Laplacian and
- growth related to parameter values.

Both phenomena are characterized by their respective timescales, *i.e.* the scale over which the phenomenon is observed in a meaningful way.

Given τ_d the diffusion timescale, its form is rather simple and can be defined as the reciprocal of the Laplacian spectral norm:

$$\tau_d\equiv\|\mathbf{L}\|_2^{-1}. \tag{14}$$

This is reasonable physically: faster diffusion is reflected in a larger Laplacian spectral norm, resulting in a smaller timescale τ_d, and *vice versa* for slower diffusion.

Conversely, given τ_g the growth timescale, its form is not unique since each model uses its own set of parameters; moreover the following assumptions are needed to define it:

G1 diffusion is ignored, that is we consider a uniform concentration in space to focus only on the growth phenomenon;

G2 the nonlinear terms are neglected, namely low concentrations[3] are studied to simplify calculations by linearizing;

G3 the growth dynamic is dominated by the longest characteristic time.

These hypotheses give the following τ_g values:

FK - applying G1–G3 to $[\mathrm{FK}]_c$, one has

$$\frac{\mathrm{d}c}{\mathrm{d}t} = \alpha c \implies c = c(0)e^{\alpha t} \implies \tau_g^{\mathrm{FK}} \equiv \alpha^{-1}. \tag{15}$$

HD - applying G1–G3 to $[\mathrm{HD}]_c$, one has

$$\begin{cases} \dfrac{\mathrm{d}c_h}{\mathrm{d}t} = \gamma - \mu_h c_h, \\ \dfrac{\mathrm{d}c_m}{\mathrm{d}t} = -\mu_m c_m, \end{cases} \implies \begin{cases} \dfrac{\mathrm{d}c_h}{\mathrm{d}t} + \mu_h c_h = \gamma, \\ c_m = c_m(0)e^{-\mu_m t}. \end{cases}$$

The first equation is a first order nonhomogeneous ODE, whose solution can be found in any basic Analysis book [4]:

$$c_h(t) = c_h(0)e^{-\mu_h t} + \frac{\gamma}{\mu_h}\left(1 - e^{-\mu_h t}\right), \tag{16}$$

while the second one has the same structure of (15); hence it's clear the growth characteristic time is

$$\tau_g^{\mathrm{HD}} \equiv \max\{\mu_h^{-1}, \mu_m^{-1}\} = \min\{\mu_h, \mu_m\}^{-1}. \tag{17}$$

SMO - applying G1–G3 to $[\mathrm{SMO}]_c$ and setting

$$\bar{\mu}_i = \mu + \frac{(i-3)}{2} f,$$

one has

$$\begin{cases} \dfrac{\mathrm{d}c_1}{\mathrm{d}t} = \gamma - \mu c_1, \\ \dfrac{\mathrm{d}c_2}{\mathrm{d}t} = -\mu c_2 + f \displaystyle\sum_{j=2}^{L-3} c_{2+j}, \\ \dfrac{\mathrm{d}c_i}{\mathrm{d}t} = -\bar{\mu}_i c_i + f \displaystyle\sum_{j=2}^{L-i-1} c_{i+j}, \; 2 < i < L, \\ \dfrac{\mathrm{d}c_L}{\mathrm{d}t} = -\mu c_L \implies c_L = c_L(0)e^{-\mu t}. \end{cases}$$

The ODE associated to c_1 has the same solution as (16), hence its characteristic time is μ^{-1}, same as the last equation. The remaining system, excluding c_1 and c_L, can be expressed as an autonomous linear system

$$\frac{\mathrm{d}\hat{\mathbf{c}}}{\mathrm{d}t} = \mathbf{A}\hat{\mathbf{c}},$$

with $\hat{\mathbf{c}} \equiv [c_2, \ldots, c_{L-1}]^\top \in \mathbb{R}^{L-2}$ and $\mathbf{A} \in \mathbb{R}^{(L-2)\times(L-2)}$ given by

$$\mathbf{A} = \begin{bmatrix} -\mu & 0 & f & \cdots & \cdots & f \\ 0 & -\bar{\mu}_3 & \ddots & \ddots & & \vdots \\ \vdots & \ddots & -\bar{\mu}_4 & \ddots & \ddots & f \\ \vdots & & \ddots & -\bar{\mu}_{L-3} & \ddots & 0 \\ \vdots & & & \ddots & -\bar{\mu}_{L-2} & 0 \\ 0 & \cdots & \cdots & \cdots & 0 & -\bar{\mu}_{L-1} \end{bmatrix},$$

whose solution is $\hat{\mathbf{c}}(t) = e^{t\mathbf{A}}\hat{\mathbf{c}}(0)$ [1] where $e^{t\mathbf{A}}$ is the matrix exponential; thus the growth characteristic time is

$$\tau_g^{\mathrm{SMO}} \equiv \max\{\mu^{-1}, |\lambda_{\mathbf{A}}^{\min}|^{-1}\} = \min\{\mu, |\lambda_{\mathbf{A}}^{\min}|\}^{-1}, \tag{18}$$

where $|\lambda_{\mathbf{A}}^{\min}|$ is the minimal absolute eigenvalue of $\mathbf{A}$.

Before continuing, by analysing dimensionally $[\mathrm{FK}]_d$, $[\mathrm{HD}]_d$ and $[\mathrm{SMO}]_d$, it's noteworthy that each quantity used in the right-hand sides of (14, 15, 17, 18) has dimension a^{-1} (for a see § 3.1); thus, τ_d and τ_g are dimensionally coherent.

With these definitions, three cases are described based on the relative value between τ_d and τ_g:

$\tau_d \ll \tau_g$: the dynamic is dominated by diffusion (Fig. 2a) and it's thus uniform in space, *i.e.*

$$\Delta^{\mathbf{D}} c = 0 \;\; \forall x \in \mathbb{R}^3 \;\text{ or }\; \mathbf{L}\mathbf{c} = 0 \text{ (see § 2.2.1)},$$

so the network behaves like a single node.

$\tau_d \approx \tau_g$: the dynamic is non-trivial (Fig. 2b), more specifically both diffusion and growth are concurrent phenomena showing biologically sound results.

$\tau_d \gg \tau_g$: the dynamic is dominated by growth (Fig. 2c) leading to a hiccup evolution in which groups of vertices are infected at different times.

This is the reason why the Laplacian has to be multiplied by ρ in (9): it's necessary in order to have $\tau_d \approx \tau_g$, where τ_g can have any of the form considered in (15, 17, 18), and thus a biologically sound dynamic.

A possible choice or ρ could be one such that $\tau_d = \tau_g$, that is

$$\tau_d = \frac{1}{\|\mathbf{L}\|_2} = \frac{1}{\rho\|\boldsymbol{\mathcal{L}}\|_2} = \tau_g \implies \rho = \frac{1}{\tau_g\|\boldsymbol{\mathcal{L}}\|_2}. \tag{19}$$

More generally, the cases illustrated before, along with Fig. 2 and Tab. 1, actually imply the presence of a continuum: while $\tau_d \ll \tau_g$ manifests very rapidly with only a difference of two orders of magnitude (OOMs), $\tau_d \gg \tau_g$ is an extreme

[3] If the initial conditions allow it, this is usually only right initially.

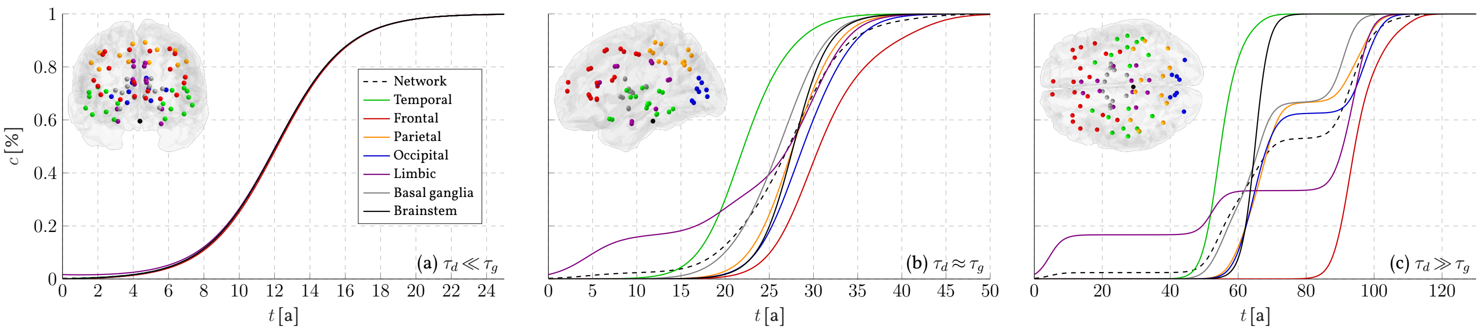

Figure 2: Three timescale cases in $[\mathrm{FK}]_d$ (see Tab. 1 for ρ, timescale and T_c data values). The plots have been generated using § 3.1.1 setting, aside Δt and T which are set appropriately in each case, and the same logic found in § 3.2.1 (*i.e.* with (56)).

case needing a difference of eight OOMs; on the contrary $\tau_d \approx \tau_g$ manifests even when there is a difference of one OOM if $\tau_d > \tau_g$, whereas if $\tau_d < \tau_g$, even with the same OOM, the evolution starts to rapidly degenerate.

Therefore, the true condition for a biologically sound dynamic is $\tau_d \gtrapprox \tau_g$ or, equivalently, for the OOMs[4]

$$\lfloor \log_{10}(\tau_d) \rfloor \geq \lfloor \log_{10}(\tau_g) \rfloor,$$

so (19) is only a *sufficient* but not *necessary* value for ρ.

	ρ	τ_d [a][1]	τ_g [a][1]	T_c [a][1,2]
$\tau_d \ll \tau_g$	1×10^{0}	$\approx 2\times 10^{-2}$	2×10^{0}	20.04
$\tau_d \approx \tau_g$	1×10^{-3}	$\approx 2\times 10^{1}$	2×10^{0}	29.69
$\tau_d \gg \tau_g$	1×10^{-10}	$\approx 2\times 10^{8}$	2×10^{0}	92.35

[1]For «a» see § 3.1. [2]For «T_c» see § 2.3.4.

Table 1: ρ, timescale and T_c data of Fig. 2.

2.2.3 Temporal discretization

The time discretization considered is the Euler method both implicit and explicit. To define it, first observe that all ODE systems in $[\mathrm{FK}]_d$, $[\mathrm{HD}]_d$ and $[\mathrm{SMO}]_d$ can be represented as

$$\frac{\mathrm{d}\mathbf{c}}{\mathrm{d}t} = \mathbf{F}(\mathbf{c}(t), t; \mathbf{L}, \mathcal{P}), \tag{20}$$

for a fixed Laplacian $\mathbf{L}$ and a set of parameters $\mathcal{P}$; of course, due to the quadratic terms, $\mathbf{F}$ is nonlinear.

Given a uniform time step size $\Delta t \in \mathbb{R}^+_*$ such that

$$T/\Delta t = N \in \mathbb{N}^+,$$

then $\mathcal{T}$ is discretized by times steps

$$t_n = n\Delta t, \text{ with } 1 \leq n \leq N.$$

The logic behind the Euler methods is to approximate the time derivative with a finite difference between two consecutive time steps:

$$\frac{\mathrm{d}\mathbf{c}}{\mathrm{d}t}(t_{n+1}) \approx \frac{\mathbf{c}(t_{n+1}) - \mathbf{c}(t_n)}{\Delta t}, \quad \forall n \in \{1, \ldots, N-1\}.$$

By evaluating $\mathbf{F}(\mathbf{c}(t), t; \mathbf{L}, \mathcal{P})$ in t_n the explicit Euler method is obtained, with update rule

$$\mathbf{c}(t_{n+1}) = \mathbf{c}(t_n) + \Delta t \mathbf{F}(\mathbf{c}(t_n), t_n; \mathbf{L}, \mathcal{P}), \tag{21}$$

while in t_{n+1} one has the implicit Euler method, with update rule

$$\mathbf{c}(t_{n+1}) = \mathbf{c}(t_n) + \Delta t \mathbf{F}(\mathbf{c}(t_{n+1}), t_{n+1}; \mathbf{L}, \mathcal{P}). \tag{22}$$

Both are first-order methods but only the second one is unconditionally stable (A-stable), which in practice means any choice of $\Delta t > 0$ results in a non-exploding solution in time whereas explicit methods need a sufficiently small time step for a non-exploding solution.

Conversely, explicit methods are less computationally demanding compared to implicit ones as the latter, due to the nonlinearity of $\mathbf{F}$, have to solve a root-finding problem for the next value. [5, 9]

Lastly, this procedure fully discretizes the solution that from (10) becomes

$$\tilde{c}(\mathbf{x}, t_n) = \sum_{i=k}^{V} c^k(t_n) \mathbb{1}_{\mathrm{ROI}_k}(\mathbf{x}) \approx c(\mathbf{x}, t_n), \tag{23}$$

with $1 \leq n \leq N$, which now lies in $\tilde{c} \in L^\infty_+(\mathcal{Q})$.

2.3 Asymptotic analyses

To normalize the plots it's necessary to find a reference value and a natural choice is the asymptotic value of the misfolded τ proteins concentration.

[4]In scientific notation $a \in \mathbb{R}^+$ is expressed as $a = b10^c$ with significand $b \in [1, 10)$ and exponent $c \in \mathbb{N}$, *i.e.* the order of magnitude; by then applying the logarithm base 10 and floor function one has, since $\log_{10}(b) \in [0, 1)$,

$$\lfloor \log_{10}(a) \rfloor = \lfloor \log_{10}(b) + c \rfloor = \lfloor \log_{10}(b) \rfloor + \lfloor c \rfloor = c.$$

Each model is studied by searching its stable equilibrium points (EPs), *i.e.* such that $\mathrm{d}/\mathrm{d}t = 0$, when the system is homogeneous, namely $\Delta^{\mathbf{D}} = 0$ or $\mathbf{L} = \mathbf{0}$.

Afterwards, an empiric formula is proposed to estimate the convergence time to reach it.

2.3.1 FK model

For $[\mathrm{FK}]_c$ one has

$$0 = \alpha c_\infty (1 - c_\infty) = f(c_\infty) \tag{24}$$

whose EPs are $c_\infty \in \{0,1\}$ as c is, in fact, already dimensionless; however from the derivative

$$f'(c_\infty) = \alpha - 2\alpha c_\infty$$

it holds

$$\begin{cases} f'(0) = \alpha > 0 \implies \text{unstable EP}, \\ f'(1) = -\alpha < 0 \implies \text{LAS EP}, \end{cases} \tag{25}$$

where LAS stands for *locally asymptotically stable* with basin of attraction $\mathcal{A} = \mathbb{R}^+_*$ (see [14] for more details): 1 is the asymptotic value of all $[\mathrm{FK}]_c$ with IC $c_0 \neq 0$, with α only changing the time required to reach it.

For $[\mathrm{FK}]_d$ one has

$$\mathbf{0} = \alpha \mathbf{c} \odot (1 - \mathbf{c}), \tag{26}$$

where each equation is like (24) but with c^i_∞, $1 \le i \le V$, in place of c_∞; hence, the asymptotic network value is $V c_\infty$.

2.3.2 HD model

For $[\mathrm{HD}]_c$ one has

$$\begin{cases} 0 = \gamma - \mu_h c_h^\infty - \kappa c_h^\infty c_m^\infty, \\ 0 = -\mu_m c_m^\infty + \kappa c_h^\infty c_m^\infty, \end{cases} \tag{27}$$

for which there are two EPs:

$$\mathbf{c}_1 \begin{cases} c_{h,1}^\infty = \dfrac{\gamma}{\mu_h}, \\ c_{m,1}^\infty = 0, \end{cases} \quad \text{and} \quad \mathbf{c}_2 \begin{cases} c_{h,2}^\infty = \dfrac{\mu_m}{\kappa}, \\ c_{m,2}^\infty = \dfrac{(\kappa\gamma - \mu_h \mu_m)}{\mu_m \kappa}. \end{cases} \tag{28}$$

From (28) it's clear $\mathbf{c}_2$ exists iff $k\gamma > \mu_h \mu_m$, otherwise it's either equal to $\mathbf{c}_1$ ($k\gamma = \mu_h\mu_m$) or $c_{m,2}^\infty$ becomes negative ($k\gamma < \mu_h \mu_m$) and thus biologically illogical.

Hence, assuming $k\gamma > \mu_h \mu_m$, it's possible, by studying the Jacobian of (27), to show (see Appendix A.2) that $\mathbf{c}_2$ in the (c_h, c_m) plane is LAS, with

$$\mathcal{A}_2 = \mathbb{R}^+ \times \mathbb{R}^+_*,$$

whereas $\mathbf{c}_1$ is unstable, though it attracts the x-axis, *i.e.*

$$\mathcal{A}_1 = \mathbb{R}^+ \times \{0\}.$$

For $[\mathrm{HD}]_d$ one has

$$\begin{cases} 0 = \gamma \mathbf{1} - \mu_h \mathbf{c}_h - \kappa \mathbf{c}_h \odot \mathbf{c}_m, \\ 0 = -\mu_m \mathbf{c}_m + \kappa \mathbf{c}_h \odot \mathbf{c}_m, \end{cases} \tag{29}$$

where for each node is described by a system like (27) but with $c_h^{i,\infty}$ and $c_m^{i,\infty}$, $1 \le i \le V$, in place of c_h^∞ and c_m^∞; thus, the asymptotic network values is V times $\mathbf{c}_1$ or $\mathbf{c}_2$.

2.3.3 SMO model

For this case it's very difficult to gain asymptotic information on the concentrations by studying the single equations as they are far too complex to handle.

However, an equivalent analysis is possible regarding the moments of the toxic aggregates

$$Q_i = \sum_{l=2}^{\infty} l^i c_l, \tag{30}$$

where only the first two are of interests:

- ◇ the total number of toxic aggregates $P \equiv Q_0$ and
- ◇ the total toxic mass $M \equiv Q_1$, expressible also as

$$M = \sum_{i=1}^{\infty} i c_i - c_1 \equiv M_{\text{tot}} - m \implies M_{\text{tot}} = m + M \tag{31}$$

  with m being the healthy monomer concentration while M_{tot} the total mass.

To gather the necessary equations the following assumption is made: the number of τ polymers L is sufficiently large as to not affect the dynamics on intermediate time scales of disease progression. [15, § 5.2] In essence it means that the sum over all equations in $[\mathrm{SMO}]_c$ is well approximated by the sum in the $L \to \infty$ limit.

Consequently, a new system of three equations can be derived by considering the first one in $[\mathrm{SMO}]_c$, with $m \equiv c_1$, the second as the sum over all equations with $i \ge 2$ in $[\mathrm{SMO}]_c$ and likewise the third but with the i-th equation multiplied by i (see Appendix B.4):

$$\begin{cases} \dfrac{\mathrm{d}m}{\mathrm{d}t} = \Delta^{\mathbf{D}} m + \gamma - \mu m - 2\boldsymbol{\kappa} m^2 - amP, & \text{(32a)} \\ \dfrac{\mathrm{d}P}{\mathrm{d}t} = \Delta^{\mathbf{D}} P - \mu P + \kappa m^2 + \dfrac{f}{2}(M - 3P + c_2), & \text{(32b)} \\ \dfrac{\mathrm{d}M}{\mathrm{d}t} = \Delta^{\mathbf{D}} M - \mu M + 2\boldsymbol{\kappa} m^2 + amP. & \text{(32c)} \end{cases}$$

Define now $m_\infty, P_\infty, M_\infty \in \mathbb{R}^+$ as the asymptotic values of the respective unknowns such that (32) become

$$\begin{cases} 0=\gamma-\mu m_\infty-2\boldsymbol{\kappa} m_\infty^2-a m_\infty P_\infty, & (33a)\\ 0=-\mu P_\infty+\kappa m_\infty^2+\dfrac{f}{2}(M_\infty-3P_\infty), & (33b)\\ 0=-\mu M_\infty+2\boldsymbol{\kappa} m_\infty^2+a m_\infty P_\infty, & (33c)\end{cases}$$

where fc_2 has been neglected since both terms are expected to be small[5], at least initially.

However, the system (33) is not yet autonomous as $\boldsymbol{\kappa}$ is not homogeneous in space due to its assumed definition in (8). The solution is to average (33) in space: all the unknowns and parameters will remain unchanged whereas $\boldsymbol{\kappa}$ becomes

$$\begin{aligned}\bar{\boldsymbol{\kappa}} &= \frac{1}{\text{Vol}(\Omega)}\int_\Omega \boldsymbol{\kappa}\,\mathrm{d}\mathbf{x} = \frac{\kappa}{\text{Vol}(\Omega)}\int_\Omega \mathbb{1}_{\mathcal{K}}\,\mathrm{d}\mathbf{x}\\ &= \frac{\kappa}{\text{Vol}(\Omega)}\int_{\mathcal{K}} 1\,\mathrm{d}\mathbf{x} = \kappa\frac{\text{Vol}(\mathcal{K})}{\text{Vol}(\Omega)} \equiv \kappa\varphi.\end{aligned} \tag{34}$$

Using (34), P_∞ is obtained by isolating it from (33a):

$$P_\infty = \frac{\gamma-\mu m_\infty-2\kappa\varphi m_\infty^2}{a m_\infty}, \tag{35}$$

which used in (33c) gives

$$M_\infty = \frac{\gamma}{\mu}-m_\infty \implies M_{\text{tot}}^\infty = M_\infty+m_\infty = \frac{\gamma}{\mu}. \tag{36}$$

This last result is also derived by finding the LAS EP of the equation obtained by summing (32a, 32c) and neglecting diffusion:

$$\frac{\mathrm{d}M_{\text{tot}}}{\mathrm{d}t} = \gamma-\mu M_{\text{tot}}, \tag{37}$$

where if $M_{\text{tot}}(0)=\gamma/\mu$ then the total mass is conserved; for this reason (36) is known as the mass conservation constrain as well.

Thereafter, using (35, 36) in (33b), one finds a third order polynomial equation

$$q_3 m_\infty^3+q_2 m_\infty^2+q_1 m_\infty+q_0=0 \tag{38}$$

with

$$\begin{aligned} &q_3=2a\kappa\varphi, && q_2=4\mu\kappa\varphi+f(6\kappa\varphi-a),\\ &q_1=2\mu^2+f[3\mu+a(\gamma/\mu)], && q_0=-\gamma(2\mu+3f).\end{aligned} \tag{39}$$

The roots of (38) can be found with Cardano's formula [8, 20], although, due to the solution complexity, their symbolic form is of little use; indeed, one then needs to use them to find M_∞ and P_∞, with (35, 36) respectively, and finally to evaluate the Jacobian of (33) to establish their nature as LAS or unstable. Therefore, the analysis of the EPs nature is omitted.

Nevertheless, in the cases studied in § 3.1, we can at least numerically say the solution satisfying the positivity constraints (see Appendix B.3) is the smallest root of (38); this result seems to be confirmed [but not proven[6]] by [15].

For $[\text{SMO}]_d$ one has to slightly change the moment definition which becomes vectorial

$$\mathbf{Q}_i = \sum_{l=2}^{\infty} l^i \mathbf{c}_l,$$

with $\mathbf{P}\equiv\mathbf{Q}_0$ and $\mathbf{M}\equiv\mathbf{Q}_1$. Then, applying to $[\text{SMO}]_d$ the same reasoning resulting in (33), leads to

$$\begin{cases} \mathbf{0}=\gamma\mathbf{1}-\mu m_\infty\mathbf{1}-2m_\infty^2\boldsymbol{\kappa}-m_\infty P_\infty\mathbf{1},\\ \mathbf{0}=-\mu P_\infty\mathbf{1}+m_\infty^2\boldsymbol{\kappa}+\dfrac{f}{2}(M_\infty\mathbf{1}-3P_\infty\mathbf{1}),\\ \mathbf{0}=-\mu M_\infty\mathbf{1}+2m_\infty^2\boldsymbol{\kappa}+m_\infty P_\infty\mathbf{1}.\end{cases}$$

Once again $\boldsymbol{\kappa}$ is non-null only in some nodes rendering the system non-autonomous and, like (34), we can solve this issue by averaging over all nodes:

$$\bar{\boldsymbol{\kappa}} = \frac{1}{V}\sum_{i=1}^{V}\kappa_i = \kappa\frac{q}{V} \equiv \kappa\varphi. \tag{40}$$

Thus the formulas for the discrete asymptotic values are the same as (35, 36, 38), but with a different φ definition and valid at each vertex, so that the asymptotic network values are V times the previous ones.

In conclusion, it's also possible to estimate the asymptotic size distribution [15, (87, 88) on p. 12] (see Appendix B.5):

$$c^\infty(n) = P_\infty K(n)\exp\left[\frac{4(f+\mu)^2-(2\mu+nf)^2}{4afm_\infty}\right], \tag{41}$$

with

$$K(n) = \frac{(2\mu+nf)^2-2afm_\infty}{4am_\infty(f+\mu)},$$

where the maximum concentration is reached by the polymer with size closest to

$$n_{\max} = \sqrt{6a\frac{m_\infty}{f}}-2\frac{\mu}{f}. \tag{42}$$

[5] This approximation is actually indistinguishable from the full system solution as can be seen in [15, Fig. 8].

[6] Indeed, in [15, § 5.5 on p. 10] they only give upper and lower bounds for m_∞ without studying the EP nature.

2.3.4 Convergence time

One might find reasonable to estimate the time necessary for the three models to reach their respective asymptotic values, based on their parameters $\mathcal{P}$.

Unfortunately this is easier said than done, in part due to the nonlinear nature of the problem at hand; nonetheless, it's natural to assume that if one does exist it must depend on the time scales found in § 2.2.2.

We here propose one such function based empirically on the cases in § 3.1:

$$T_c = 10\tau_g f(\tau_d/\tau_g) = 10\tau_g \left[\frac{9}{20}\log_{10}\left(\frac{\tau_d}{\tau_g}+1\right)+1\right], \quad (43)$$

from which the following remarks can be made:

R1 if $\tau_d \ll \tau_g$ then the network behaves like one node (§ 2.2.2) and $f(\tau_d) \approx 1$;

R2 if $\tau_d > \tau_g$ then diffusion is going to slow down convergence but only slightly as only a small quantity is needed to spark growth;

R3 $9/20$ was chosen as to almost double $f(\tau_d/\tau_g)$ when $\log_{10}(\tau_d/\tau_g+1)$ doubles as well[7];

R4 10 comes from the observations that the convergence time of a single node is tenfold the initial estimated time;

R5 as § 2.2.2, (43) is *not* a precise estimation of the convergence time, but it must be rather seen as an approximation of the OOM, *i.e.* $\lfloor\log_{10}(T_c)\rfloor$, necessary to asymptotically converge.

Assumption R2 is motivated with the following reasoning: from § 2.3.1–2.3.3 it's clear the only LAS EPs are those with a non-null toxic concentration; hence, if

$$1 \ll \tau_d/\tau_g < +\infty,$$

Lc can be seen as the application of small perturbations to healthy nodes until their state changes from the unstable healthy one[8]; and since the size of the perturbation is less important than its presence, a slow growth must be considered to account for τ_d/τ_g influence to T_c.

As a last note, we remark that there definitely exists a better formula for T_c than (43) (for instance the constant $9/20$ can be ameliorated), though we are quite certain about the logarithmic growth inside $f(\tau_d/\tau_g)$.

[7] For instance, if τ_d/τ_g goes from 10^{10} to 10^{20}, doubling $\log_{10}(\tau_d/\tau_g+1)$, then $f(\tau_d/\tau_g)$ almost doubles from 5.5 to 10.

[8] That is to say that, no matter how feeble diffusion may be and as long as all nodes are connected by at least a non-null edge, the network is going to be completely infected, exactly like its continuous counterpart.

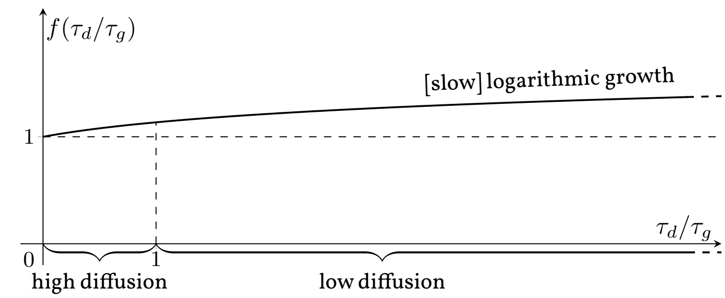

Figure 3: Plot of $f(\tau_d/\tau_g)$ from (43).

2.4 Complications

Modifications are needed to make the models more complex and thus more realistic. In total only three are considered: one for all models and two for the SMO one.

2.4.1 Dynamic Laplacian

Ageing is a natural and inevitable process in all our lives, and one crucial way it manifests is in the weakening of the body. Thus, the fact that the Laplacian in (9) does not depend on time is in fact a mere simplification.

But how would the brain depend on time? The answer is not easy and instead of finding it in the literature [if there exists one] we propose two possible behaviours:

B1 either all edges decay with the same rate in time

B2 or edges decay w.r.t. some quantity: the bigger it is, the faster/slower they decay in time.

These reflect how the brain will react in time: if it tries to preserve its form, deeming it optimal, then B1 is right; otherwise it will try to maintain certain axons over others based on some quantity like in B2.

Between the two B2 is studied simply because it's the most interesting from a theoretical point of view; indeed B1 will simply weaken diffusion as (14) increases.

As the control quantity one might propose the distance, but due to the high anisotropy of the brain this is a poor choice. A better one are the entries of the adjacency matrix **A** (Fig. 1), which are defined as the ratio of the number of fibres over the average fibre length: n/l [2, 17, 18]; therefore, one can arbitrarily assume that the larger A_{ij} is, the greater the brain will try to maintain its associated edge.

To classify the entries as low/medium/high value the maximum $A_{\text{max}} \approx 9.816$, minimum $A_{\text{min}} = 0$ and average

$$\overline{A} \equiv \frac{1}{V^2}\sum_{i,j}^{V} A_{ij} \approx 0.134$$

values of **A** are considered; then the midpoints between

$\overline{A}$-$A_{\max}$ and $\overline{A}$-$A_{\min}$ defines the thresholds needed to classify the entries:

$$\begin{aligned} S_{lm} &\equiv \frac{A_{\min}+\overline{A}}{2} = 6.72\times 10^{-2}, \\ S_{mh} &\equiv \frac{\overline{A}+A_{\max}}{2} \approx 4.975, \end{aligned} \tag{44}$$

as illustrated in Fig. 4.

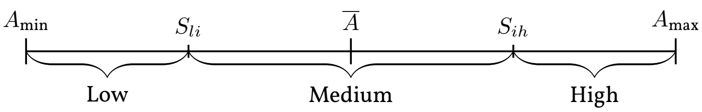

Figure 4: Classifying thresholds.

In total the number of low/medium/high value edges are 5785, 1094 and 10 respectively, which sum to 83^2 as expected. With all this, $\mathbf{A}(t)$ is defined as follows

$$A_{ij}(t) \equiv \begin{cases} A_{ij} f_d^l(t) & \text{if } A_{\min} \le A_{ij} < S_{lm} \\ A_{ij} f_d^m(t) & \text{if } S_{lm} \le A_{ij} < S_{mh} \\ A_{ij} f_d^h(t) & \text{if } S_{mh} \le A_{ij} < A_{\max} \end{cases} \tag{45}$$

where the three decaying functions are

$$f_d^l(t) \equiv \exp\left[-\frac{t}{2}\right], \quad f_d^m(t) \equiv \frac{1}{1+\frac{t}{10}}, \quad f_d^h(t) \equiv \frac{1}{1+\frac{t}{40}}, \tag{46}$$

shown in Fig. 5.

Afterwards, $\mathbf{L}$ is defined exactly as (9):

$$\mathbf{L}(t) \equiv \rho(\mathbf{D}(t)-\mathbf{A}(t)) = \rho\boldsymbol{\mathcal{L}}(t), \tag{47}$$

but with $\mathbf{D}(t) \equiv \operatorname{diag}(\mathbf{A}(t)\mathbf{1})$.

It's noteworthy to emphasize that the kinematics described by (45) is purely theoretical and based on the assumption that the low value edges decay very rapidly (exponential decay), whereas those medium and high value are more resistant (hyperbolic decay).

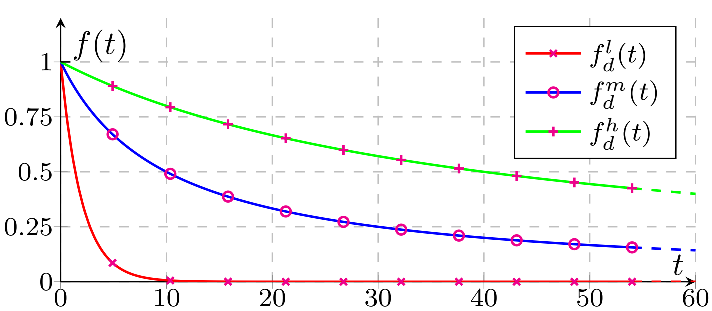

Figure 5: Decay functions.

2.4.2 Accumulation

In $[\mathrm{SMO}]_c$ it's possible to assume the last L-th polymer cannot be cleared and accumulate in time, leading to brain damages; this can be taken into account by setting $\mu_L = 0$ instead of $\mu_L = \mu$ in A4.

2.4.3 Size-dependency

Still within $[\mathrm{SMO}]_c$, it's possible to assume that both diffusion and clearance rate depend on the polymer size; this is sensible as the larger a polymer is, the less is able to diffuse and the longer it takes to break it down.

For diffusion, A2 can be changed so that the diffusion coefficient $\mathbf{D}_i$ of the i-th polymer scales approximately as a power of its molecular weight which is proportional to the polymer size: [15, (23) on p. 5]

$$\mathbf{D}_i \equiv i^{-\eta}\mathbf{D} \quad \Longrightarrow \quad \mathbf{L}_i \equiv i^{-\eta}\mathbf{L}, \tag{48}$$

where $\eta = 0.5$.

For clearance, A4 can be modified by setting $\mu_i = \mu/i$; in this way, the larger the polymer is, the slower it's cleared. This last change actually breaks mass conservation in (37) which becomes [15, (47) on p. 8]

$$\frac{\mathrm{d}M_{\text{tot}}}{\mathrm{d}t} = \gamma - \mu P_{\text{tot}}, \tag{49}$$

where $P_{\text{tot}} \equiv \sum_{i=1}^{L} c_i = c_1 + P = m + P$, so mass is expected to grow even if $M_{\text{tot}}(0) = \gamma/\mu$. The same result [but not (49)] is true, to a lesser extent, with the previous complication in which only $\mu_L = 0$ is different.

3 Settings and results

In this section the parameters and initial conditions of the models are set, then the results are discussed by comparing the base models with and without complications.

3.1 Settings

In few words, all parameters come from [16]; $[\mathrm{SMO}]_d$ has been tested with those taken from [15] as well. In particular, two settings are shared between all models:

- the infected nodes, *i.e.* where the initial toxic conditions are set (§ 2.2), are those in the entorhinal cortex ($q = 2$) in the Limbic system, shown in Fig. 6;
- ρ is chosen such that

$$\rho \equiv \|\mathbf{L}\|_2^{-1} = 0.0219 \quad \Longrightarrow \quad \tau_d = 1. \tag{50}$$

In conclusion to this section, some remarks are necessary in regard to all parameters.

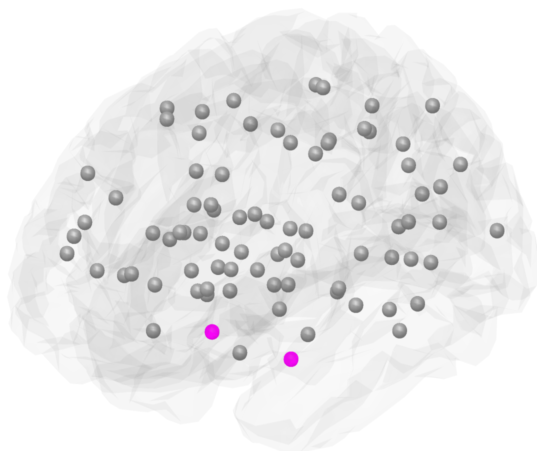

Figure 6: Entorhinal infected nodes coloured magenta.

3.1.1 FK model

The $[\text{FK}]_d$ settings are the following:

- the parameter takes value
$$\alpha = 0.5\text{a}^{-1}, \tag{51}$$
where a stands for *annus*, meaning year in Latin. Of course, [tropical] years aren't a fundamental dimension, although they can be converted in seconds, as suggested by the SI [7, 19], with
$$1\,\text{a} \approx 365.24220\,\text{d} \approx 3.1556926 \times 10^7\,\text{s}; \tag{52}$$
- τ_g and T_c values, from (15, 43) respectively, can be found in Tab. 2;
- the initial toxic concentration is $c_0 = 0.1$;
- the Euler implicit method in (22) is considered with $\Delta t = 0.4\text{a}$ and $T = 30\text{a}$.

3.1.2 HD model

The $[\text{HD}]_d$ settings are the following:

- the parameters are
$$\begin{aligned} \gamma &= 1\text{M}\cdot\text{a}^{-1}, & \kappa &= 0.5\text{M}^{-1}\cdot\text{a}^{-1}, \\ \mu_h &= 0.5\text{a}^{-1}, & \mu_m &= 0.5\text{a}^{-1}, \end{aligned} \tag{53}$$
where M is molar concentration defined in SI units as [11]
$$\text{M} = \text{mol}/\text{m}^3;$$
- τ_g and T_c values, from (17, 43) respectively, can be found in Tab. 2;
- the asymptotic values from (28) are
$$c_{h,2}^{\infty} = 1\text{M}, \quad \text{and} \quad c_{m,2}^{\infty} = 1\text{M};$$

	ρ	τ_d [a]	τ_g [a]	T_c[a]
$[\text{FK}]_d$	2.19×10^{-2}	1×10^{0}	2×10^{0}	21.58
$[\text{HD}]_d$	2.19×10^{-2}	1×10^{0}	2×10^{0}	21.58
$[\text{SMO}]_d$	2.19×10^{-2}	1×10^{0}	2×10^{0}	21.58

Table 2: Constant ρ, timescales and convergence time of the case studies.

- the initial healthy concentration is $c_{h,0} = 2\text{M}$ while the initial toxic one is $c_{m,0} = 0.1\text{M}$;
- the Euler implicit method in (22) is considered with $\Delta t = 0.4\text{a}$ and $T = 30\text{a}$.

3.1.3 SMO model

The $[\text{SMO}]_d$ settings are the following:

- the parameters are
$$\begin{aligned} \gamma &= 1\text{M}\cdot\text{a}^{-1}, & \mu &= 0.5\text{a}^{-1}, \\ a &= 10\text{M}^{-1}\cdot\text{a}^{-1}, & f &= 0.096\text{a}^{-1}, \\ \kappa &= 1.6\times10^{-4}\text{M}^{-1}\cdot\text{a}^{-1}, & L &= 50; \end{aligned} \tag{54}$$
- τ_g and T_c values, from (18, 43) respectively, can be found in Tab. 2;
- numerically the asymptotic values from (35, 36, 38) (with φ from (40)) are
$$\begin{aligned} m_\infty &= 0.6708\text{M}, \\ M_\infty &= 1.3292\text{M}, \\ P_\infty &= 0.0991\text{M}; \end{aligned}$$
- the initial healthy concentration is $c_{1,0} = 2\text{M} = \gamma/\mu$;
- the Euler explicit method in (21) is considered with $\Delta t = 0.04\text{a}$, to ensure its stability, and $T = 60\text{a}$.

3.1.4 Disclaimer

The data in [16] have three majors problems:

P1 the authors don't mention their dimensions which from the plots we assumed to be in a;

P2 the authors don't reference any studies from which their parameters are taken;

P3 the authors do not provide the Laplacian used.

To solve P3 we could build the Laplacian or take another already built; as written in § 2.2.1, we have chosen to follow the latter by considering the one in [15], which is, however, not provided directly in the publication but in its preprint [6].

Instead, P2 suggests the parameters are purely theoretical which seems to be the case even in [15]; in both papers, however, this is only implied by context and not explicitly written. For this reason we advise considering all previous parameters as merely theoretical.

It's then to no-one's surprise the results in [16] have unfortunately not been reproduced correctly:

- ◇ the FK and HD simulations have the same convergence time but they manifest a different dynamic in which their spreading is clockwise:

$$\text{temporal} \to \text{frontal} \to \text{parietal} \to \text{occipital}, \tag{cw}$$

  starting from the entorhinal cortex, whereas ours is anticlockwise, *i.e.* it reverses the last three regions:

$$\text{temporal} \to \text{occipital} \to \text{parietal} \to \text{frontal}. \tag{acw}$$

- ◇ the SMO simulation has both a different spreading and convergence time about 30 a slower.

Nevertheless, these discrepancies are not crucial in this paper as we are only interested in comparing the base and modified models, which is possible even only on theoretical grounds.

3.2 Results

The simulation results using the parameters of the previous section are discussed first for $[\mathrm{FK}]_d$ and $[\mathrm{HD}]_d$ at once, and then for $[\mathrm{SMO}]_d$.

All simulations have been carried out with MATLAB software version «24.2.0.2923080 (R2024b) Update 6», alongside Optimization Toolbox version 24.2.

3.2.1 FK and HD model

Instead of showing the concentration of all nodes, it's better to visually pool some together and in both models the group of ROIs considered form

- ◇ either a system (limbic, basal ganglia),
- ◇ a lobe (temporal, occipital, parietal or frontal),
- ◇ a single node (brainstem) or
- ◇ the whole network.

Given $\mathcal{G}$ the set of $|\mathcal{G}|$ nodes in one of the previous groups, then its normalized concentration is defined as

$$c_{\mathcal{G}} \equiv \frac{1}{\mathcal{N}|\mathcal{G}|}\sum_{v\in\mathcal{G}} c^v, \tag{56}$$

where $\mathcal{N}$ is the normalization constant: 1 for $[\mathrm{FK}]_d$ (§ 2.3.1) and $c^{\infty}_{m,2}$ in (28) for $[\mathrm{HD}]_d$ (§ 2.3.2). The resulting concentrations are shown in Fig. 8 for $[\mathrm{FK}]_d$ and Fig. 9 for $[\mathrm{HD}]_d$.

As can be seen qualitatively in Fig. 7 for the network, $[\mathrm{FK}]_d$ and $[\mathrm{HD}]_d$ are very similar with the former slightly underestimating the latter by an OOM of 10^{-3}. This is the reason why they are discussed together in this section.

This result is actually quite interesting: if one were to transform $[\mathrm{HD}]_d$ in $[\mathrm{FK}]_d$ (see Appendix A.1) then the α from (A3) with (53) is exactly 0.5, the same as (51); it is, however, not true that $c_m^{\max}$ from (A3) is the maximum concentration of Fig. 8 as the condition $\kappa/\mu_h \ll 1$ is not satisfied. In addition, a more rigorous analytical proof[9] is needed to truly prove this equivalence, but, since it's beyond the scope of the present paper, it is ignored; at any rate, if true, it would mean that the normalized [w.r.t. $c^{\infty}_{m,2}$ in (28)] curves of $[\mathrm{HD}]_d$ can be calculated with the simpler $[\mathrm{FK}]_d$, discarding V unknowns.

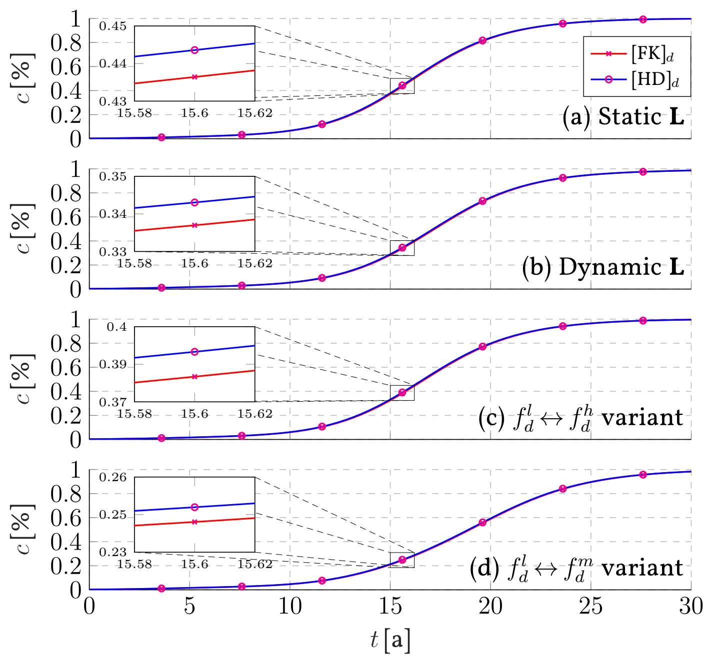

Figure 7: $[\mathrm{FK}]_d$ and $[\mathrm{HD}]_d$ network comparison.

The static **L** case is shown in Figs. 8a and 9a; in this the (acw) spreading manifests, whereas the fact the parietal and occipital lobe have overlapping curves is a clear sign of the degeneracy onset (§ 2.2.2) and, indeed, $\tau_d < \tau_g$.

Conversely, the dynamic **L** case is illustrated in Figs. 8b and 9b where the parietal and occipital lobes do not have overlapping curves any more. Meanwhile, from Figs. 8e–8l and 9e–9l it's clear the presence of a slight global delay, that is to say that at one time the concentration is less than in **L** static case.

Nevertheless, overall the difference between the static and dynamic **L** cases is very minimal considering all 5785

[9] The authors in [16, § 4.2] suggests the same conclusion but do not prove it, opting instead to discuss some parametric studies of $[\mathrm{FK}]_d$ and $[\mathrm{HD}]_d$ [unfortunately without normalizing the latter].

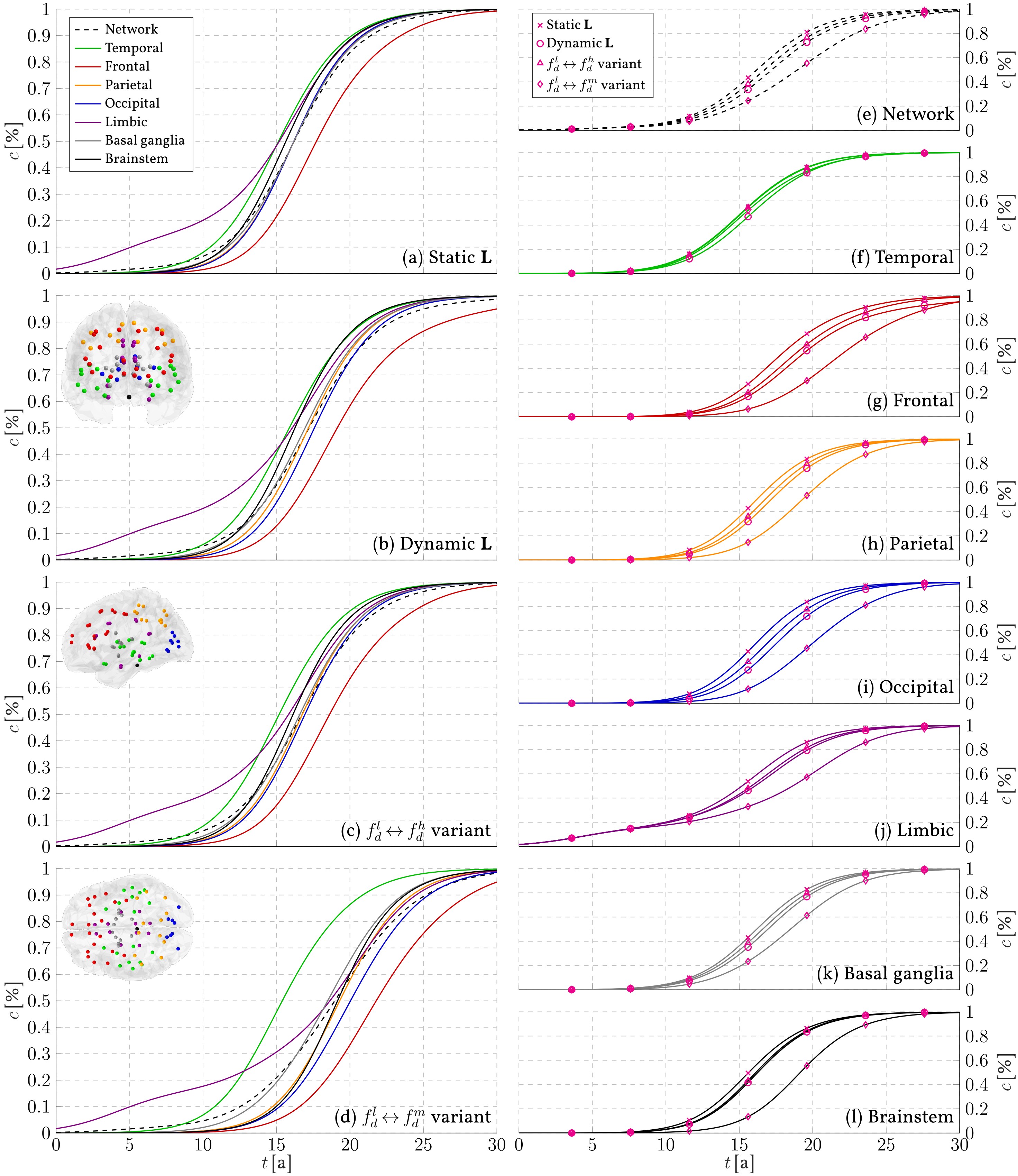

Figure 8: $[\mathrm{FK}]_d$ case studies.

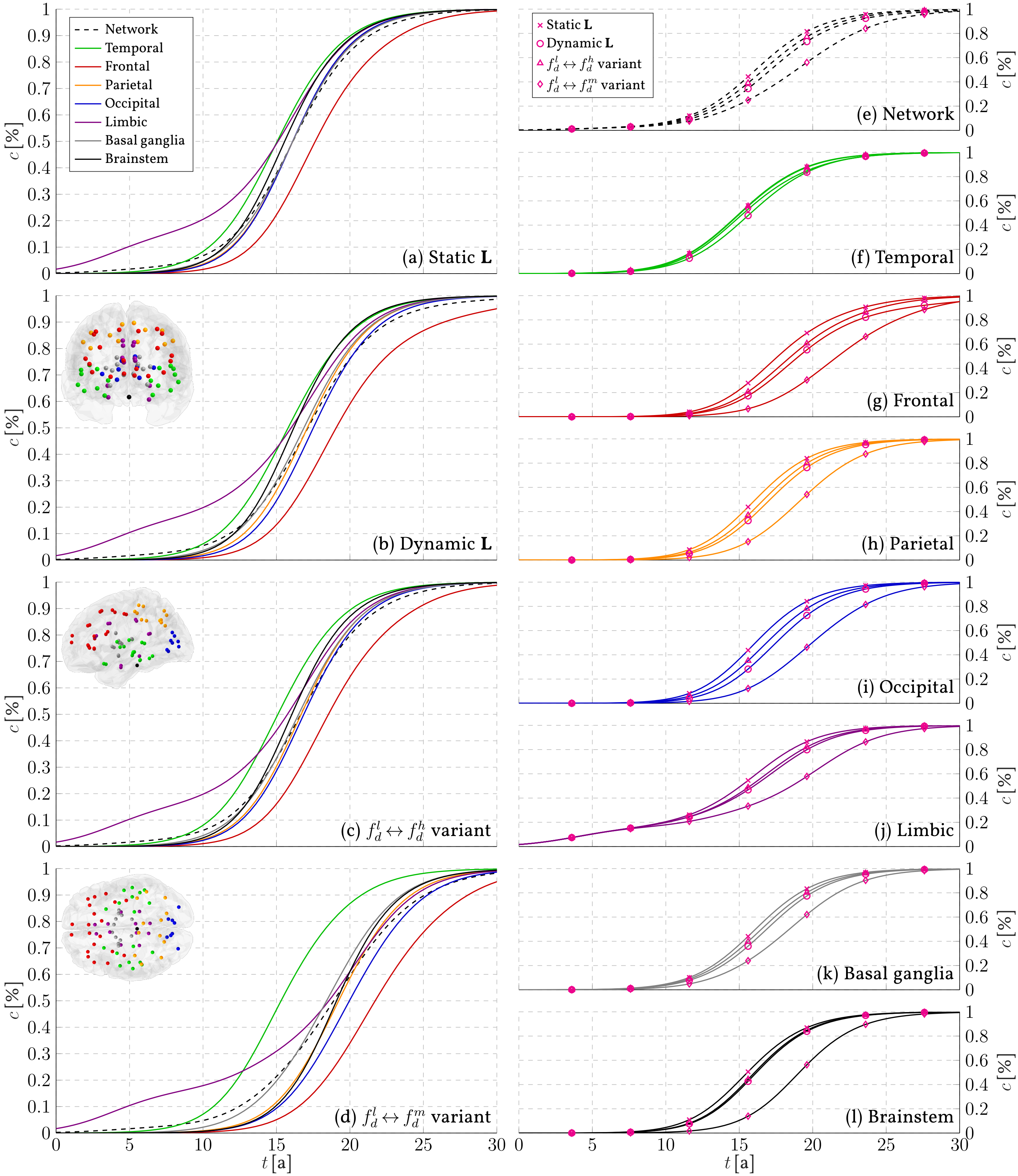

Figure 9: $[\mathrm{HD}]_d$ case studies.

low value edges die out in about 1/3 of the dynamic: this implies their influence is small. To test this intuition we have proposed two variants: one where f_d^l is swapped with f_d^h and the other with f_d^m, represented with

$$f_d^l \leftrightarrow f_d^h, \tag{V1}$$
$$f_d^l \leftrightarrow f_d^m. \tag{V2}$$

The goal is to see how the network behaves if the medium and high value edges die out very early in the dynamic.

The (V1) case can be seen in Figs. 8c and 9c and are very similar to the dynamic **L** case. From Figs. 8e–8l and 9e–9l it's evidently a middle way between the static and dynamic **L** cases. Consequently, the few 10 high value edges are not really relevant for the diffusion phenomenon.

On the contrary, the (V2) case is shown in Figs. 8d and 9d and appears to be very different: the basal ganglia outgrow the brainstem concentration, while the Limbic system slows down and is outgrown by the parietal lobe around the end. From Figs. 8e–8l and 9e–9l, instead, it's clear this is by far the most delayed dynamic of all, aside from the temporal lobe.

Therefore, the (V2) case hints at the importance of the 1094 medium value edges for the dynamic of misfolded τ proteins. One could attribute this behaviour to the bigger increase of the diffusive timescale τ_d, but Fig. 10 proves this may not the case: the (V2) τ_d increases more in the first half but is then surpassed in value by the (V1) τ_d; at any rate, they still remain in the same OOM of τ_g, far from the hiccup evolution described in § 2.2.2.

Lastly, it's noteworthy that the convergence time increases only slightly in all cases, confirming R2.

3.2.2 SMO model

Contrary to before, in this case it's better to take a more holistic approach[10] where the specific groups are abandoned in favour of the normalized aggregate masses, defined as

$$\mathrm{M}_n \equiv \frac{1}{V M_\infty} \sum_{v=1}^{V} \sum_{l=n}^{L} l c_l^v, \quad 2 \le n \le L, \tag{58}$$

and the normalized [w.r.t. P_∞] size distribution, *i.e.* the concentration against the polymer size. The resulting plots are shown in Fig. 11.

The static **L** case is illustrated in Figs. 11a and 11e. The only thing to note is that the size distribution is pretty close to the asymptotic estimate, and this is true even for the average polymer length defined as the weighted mean

[10] For this reason the variants discussed for the FK and HD models are omitted in this section.

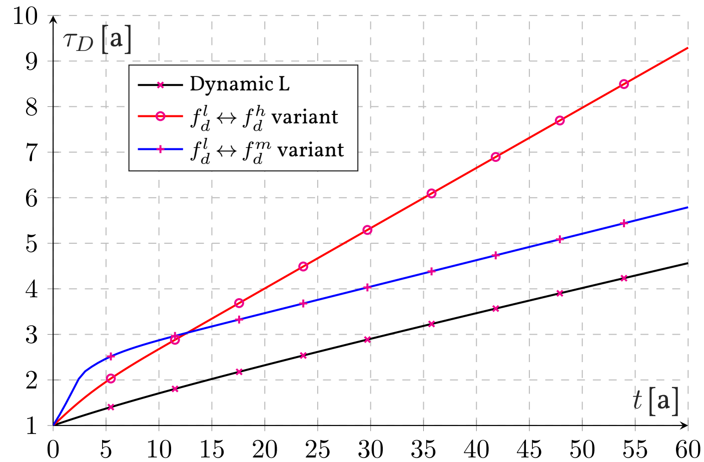

Figure 10: Dynamic diffusive timescale.

of the sizes w.r.t. their concentrations:

$$\bar{l} \equiv \frac{\sum_{v=1}^{V} \sum_{l=1}^{L} l c_l^v}{\sum_{v=1}^{V} \sum_{l=1}^{L} c_l^v} \overset{(*)}{=} \frac{\cancel{V} \sum_{l=1}^{L} l c_l}{\cancel{V} \sum_{l=1}^{L} c_l} = \frac{M_\infty}{P_\infty}, \tag{59}$$

where $(*)$ is only valid asymptotically when all nodes have the same concentration. It's noteworthy that the average length of the asymptotic size distribution estimate is

$$\lambda \approx 13.22.$$

As expected, in the dynamic **L** case from Fig. 11b it's clear the delayed nature of the aggregate masses compared to Fig. 11a; this is very slightly reflected in the size distribution in Fig. 11f as well, since it's smaller than Fig. 11e and it has a greater average length than λ.

Conversely, the absence of clearance of the last polymer shows its accumulation pretty well with both the presence of a toxic mass base in Fig. 11c, while in Fig. 11g the most noticeable difference is the spike in $n = 50$ which shifts the average length to a greater value w.r.t. λ, though overall the distribution remains the same as Fig. 11e. Observe how this spike is present even in Figs. 11e and 11f, but only temporarily, due to the absence of aggregation processes for the longest toxic polymer in $[\mathrm{SMO}]_d$.

Finally, the size dependency has the most drastic effects with an eight-fold increase of M_2 in Fig. 11d and a five-fold one of the size distribution in Fig. 11h; furthermore, the major growth happens around 10 a before the static **L** case, certainly because of the lower clearance rate for all polymers. It's then to no-one's surprise the asymptotic size distribution is completely wrong. Nevertheless, it was expected from (49) as the new EP is now in $P_{\text{tot}} = \gamma/\mu$; indeed, by plotting P_{tot} in time, as shown in Fig. 12, one finds $P_{\text{tot}} < \gamma/\mu$, meaning $\dot{M}_{\text{tot}} > 0$, for 60 a.

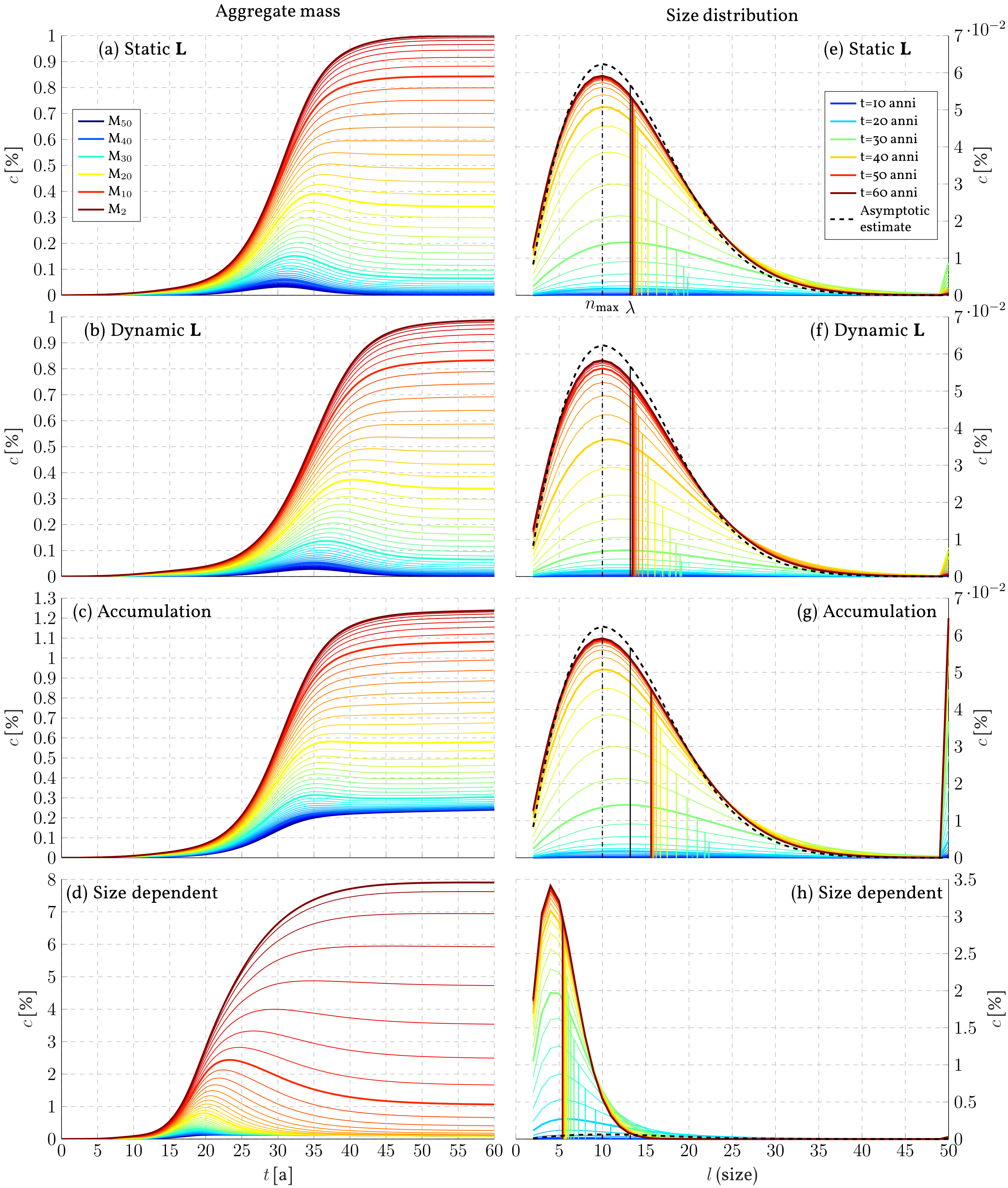

Figure 11: $[\mathrm{SMO}]_d$ case studies.

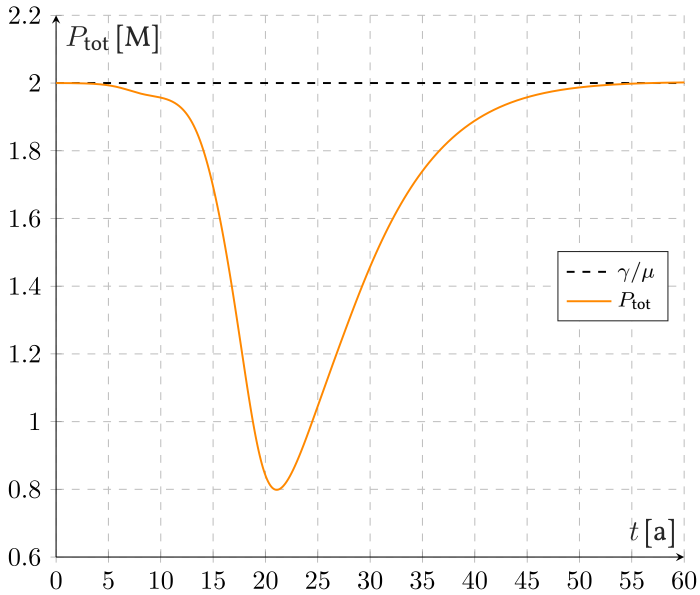

Figure 12: P_{tot} evolution in size-dependency case.

4 Conclusions

All results in § 3.2 are coherent with the theory explained in § 2, except for two:

- in the $f_d^l \leftrightarrow f_d^m$ variant for $[\text{FK}]_d$ and $[\text{HD}]_d$, it's not clear why the medium value edges are so impactful on the dynamic, even though they are not the majority; a possible motivation comes from how the graph topology changes: it's reasonable to assume it's basically unaffected in the other cases, whereas in the $f_d^l \leftrightarrow f_d^m$ variant it varies significantly.
- In the size dependent case for $[\text{SMO}]_d$, one could expect the toxic mass to increase indefinitely, particularly due to the null-value clearance rate for large enough polymers: indeed, if the toxic polymers keep growing in size, P_{tot} can remain under γ/μ while M_{tot} increases; still, a more rigorous proof is needed to verify why this is not the case, maybe akin to the moment analysis discussed in § 2.3.3.

Nevertheless, as with most simulations, they have to be taken carefully; this is true especially from a biomedical point of view since the parameters used are purely theoretical and do not come from any study (§ 3.1.4).

Furthermore, to confirm the Laplacian evolution in § 2.4.1, it has to be compared with actual measurements in vivo, which is understandably difficult to achieve. It would also be interesting to find some laws about its evolution, either based on ageing or misfolded τ protein concentrations, and hence their damage done to the brain.

Another possible improvement can be searched in parameters varying both in time and space, but once again the problem is to find their appropriate evolution. Indeed, for instance, one could expect in a diseased brain to have a nucleation and a healthy clearance rate respectively increasing/decreasing in time. Moreover, perhaps, their evolution is slow enough to justify their current constant value with an average.

All considered, this paper is better to be regarded as a formal exploration of more complex models and their overall phenomenology.

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Appendix

A HD model

A.1 $[\mathrm{FK}]_c$ derivation from $[\mathrm{HD}]_c$

Assuming that in $[\mathrm{HD}]_c$ the healthy protein is both stationary, $\mathrm{d}c_h/\mathrm{d}t = 0$, and homogeneous, $\Delta^{\mathbf{D}} c_h = 0$, then one has

$$0 = \gamma - \mu_h c_h - \kappa c_h c_m, \tag{A1}$$

which, for instance, might happen if $c_h \gg c_h c_m$, that is $c_m \ll 1$, so that the reaction term negligible with respect to the clearance one.

Isolating c_h, one has

$$c_h = \frac{\gamma}{\mu_h + \kappa c_m} = \frac{\gamma}{\mu_h} \frac{1}{1 + \frac{\kappa}{\mu_h} c_m} \overset{(*)}{\simeq} \frac{\gamma}{\mu_h}\left(1 - \frac{\kappa}{\mu_h} c_m\right),$$

where in $(*)$ it was assumed $\kappa/\mu_h = \varepsilon \ll 1$ which justifies the use of Taylor expansion. [4]

Now, using (A.1) in the c_m equation, it holds

$$\begin{aligned}
\frac{\mathrm{d}c_m}{\mathrm{d}t} &= \Delta^{\mathbf{D}} c_m - \mu_m c_m + \kappa c_m \frac{\gamma}{\mu_h}\left(1 - \frac{\kappa}{\mu_h} c_m\right) \\
&= \Delta^{\mathbf{D}} c_m + c_m \left[\frac{\kappa\gamma - \mu_m \mu_h}{\mu_h} - \gamma \frac{\kappa^2}{\mu_h^2} c_m\right] \\
&= \Delta^{\mathbf{D}} c_m + \frac{\kappa\gamma - \mu_m \mu_h}{\mu_h} c_m \left[1 - \frac{\gamma\kappa^2}{(\kappa\gamma - \mu_m\mu_h)\mu_h} c_m\right].
\end{aligned} \tag{A2}$$

Finally, by setting

$$\alpha \equiv \frac{\kappa\gamma - \mu_m \mu_h}{\mu_h} \quad \text{and} \quad c_m^{\max} \equiv \frac{(\kappa\gamma - \mu_m \mu_h)\mu_h}{\gamma\kappa^2}, \tag{A3}$$

and making the concentration dimensionless with respect to $c_m^{\max}$, *i.e.* $c_m = c_m^{\max} c$, the FK model is finally obtained:

$$\frac{\mathrm{d}c}{\mathrm{d}t} = \Delta^{\mathbf{D}} c + \alpha c(1-c). \tag{A4}$$

A.2 Asymptotic $[\mathrm{HD}]_c$ calculations

The Jacobian of (28) is

$$J(\mathbf{c}) \equiv J(c_h, c_m) = \begin{pmatrix} -\mu_h - \kappa c_m^\infty & -\kappa c_h^\infty \\ \kappa c_m^\infty & -\mu_m + \kappa c_h^\infty \end{pmatrix}. \tag{A5}$$

Evaluating it in $\mathbf{c}_1$ gives, setting $J_1 \equiv J(\mathbf{c}_1)$,

$$J_1 = \begin{pmatrix} -\mu_h & -\kappa \frac{\gamma}{\mu_h} \\ 0 & -\mu_m + \kappa \frac{\gamma}{\mu_h} \end{pmatrix}, \tag{A6}$$

with trace and determinant

$$\begin{cases} \mathrm{tr}(J_1) = -\mu_h - \mu_m + \kappa \dfrac{\gamma}{\mu_h}, \\ \det(J_1) = \mu_h \mu_m - \kappa\gamma. \end{cases} \tag{A7}$$

From these results, it's clear that if $\det(J_1) < 0$ then $\mathbf{c}_1$ is definitely unstable, while if $\det(J_1) > 0$ the trace is negative,

$$\mu_h \mathrm{tr}(J_1) = -\mu_h^2 - \det(J_1) < 0 \overset{\mu_h > 0}{\Longrightarrow} \mathrm{tr}(J_1) < 0,$$

meaning the point is LAS.

Instead, evaluating (A5) in $\mathbf{c}_2$ one has, setting $J_2 \equiv J(\mathbf{c}_2)$,

$$J_2 = \begin{pmatrix} -\frac{\kappa\gamma}{\mu_h} & -\mu_m \\ \frac{\kappa\gamma - \mu_h\mu_m}{\mu_m} & 0 \end{pmatrix}, \tag{A8}$$

with trace and determinant

$$\begin{cases} \mathrm{tr}(J_2) = -\dfrac{\kappa\gamma}{\mu_h} < 0, \\ \det(J_2) = \kappa\gamma - \mu_h\mu_m = -\det(J_1). \end{cases} \tag{A9}$$

Since $\mathrm{tr}(J_2)$, $\mathbf{c}_2$ stability is determined entirely by its deteriminant: if $\det(J_2) > 0$ then [it exists and] it's LAS, otherwise [it doesn't exists and] it's unstable. Moreover, from $\det(J_2) = -\det(J_1)$ we can deduce it's behaviour is opposita that of $\mathbf{c}_1$: if one is LAS the other is unstable and *vice versa*.

Lastly if $\kappa\gamma = \mu_h \mu_m$ then $\mathbf{c}_1 = \mathbf{c}_2$ and $\det(J_i) = 0$, $i = 1,2$, meaning it's a critical EP [14] whose nature is more difficult to study; indeed the Jacobian is in this case of no use since one eigenvalue is null. However, as we are interested only in cases in which $c_m^\infty > 0$, the study of this very specific case is omitted.

B SMO model

B.1 Nondimensionalization

The FK and HD models are studied without introducing further nondimensionalization; thus the FK one is semidimensional as the parameters are dimensional, but the concentration is not, while the HD one is fully dimensional.

[15] consider a version $[\mathrm{SMO}]_c$ which is dimensionless.

On the contrary the SMO model is nondimensionalized following the logic in § 3.3 on p. 6 of [15]; therefore, given m_0 the initial [non-zero] healthy monomer concentration the dimensionless parameters are

$$\begin{aligned} &\tilde{a} = \frac{a}{a} = 1, && \tilde{f} = \frac{f}{a m_0}, && \tilde{\gamma} = \frac{\gamma}{a m_0^2}, \\ &\tilde{\mu} = \frac{\mu}{a m_0}, && \tilde{\kappa} = \frac{\kappa}{a}, && \tilde{\mathbf{D}} = \frac{\mathbf{D}}{a m_0}, \end{aligned} \tag{B1}$$

while the dimensionless concentrations and time are

$$\tilde{c}_i = \frac{c_i}{m_0} \quad \text{and} \quad \tilde{t} = (am_0)t. \tag{B2}$$

Thanks to (B1, B2), $[\mathrm{SMO}]_c$ becomes, dropping the tildes,

$$\begin{cases} \left.\begin{aligned} &\frac{\mathrm{d}c_1}{\mathrm{d}t} = \Delta^{\mathbf{D}} c_1 + \gamma - \mu c_1 - 2\boldsymbol{\kappa} c_1^2 - c_1 \sum_{j=2}^{L-1} c_j, \\ &\frac{\mathrm{d}c_2}{\mathrm{d}t} = \Delta^{\mathbf{D}} c_2 - \mu c_2 + \boldsymbol{\kappa} c_1^2 - c_1 c_2 + f \sum_{j=2}^{L-3} c_{2+j}, \\ &\frac{\mathrm{d}c_i}{\mathrm{d}t} = \Delta^{\mathbf{D}} c_i - \mu c_i + c_1 c_{i-1} - c_1 c_i \\ &\qquad - \frac{(i-3)}{2} f c_i + f \sum_{j=2}^{L-i-1} c_{i+j}, \; 2<i<L \\ &\frac{\mathrm{d}c_L}{\mathrm{d}t} = \Delta^{\mathbf{D}} c_L - \mu c_L + c_1 c_{L-1}. \end{aligned} \right\} \text{in } \mathcal{Q}, \\ \text{BC: } c_i = 0, \; 1 \le i \le L, \qquad \text{on } \partial\mathcal{Q}_x, \\ \text{IC: } c_1 = c_{1,0}, \, c_i = 0, \; 1 < i \le L, \qquad \text{on } \partial\mathcal{Q}_t, \end{cases} \tag{B3}$$

B.2 Euler contracted methods

For (B3) usually the dimensionless duration T is much greater than Δt leading to an excessive number of time steps $N \gg 1$ slowing down the simulation.

One possible way to avoid such a fate, and to process the same results in a much faster way, is to contract the time duration; take $\varepsilon \ll 1$ such that

$$t^c = t\varepsilon, \quad T^c \equiv \varepsilon T \ll T, \quad \text{and} \quad N^c \equiv T^c/\Delta t \ll N,$$

then (20), given

$$\frac{\mathrm{d}}{\mathrm{d}t} = \varepsilon \frac{\mathrm{d}}{\mathrm{d}t^c},$$

becomes

$$\varepsilon \frac{\mathrm{d}\mathbf{c}}{\mathrm{d}t^c} = \mathbf{F}(\mathbf{c}(t^c/\varepsilon), t^c/\varepsilon; \mathbf{L}, \mathcal{P}). \tag{B4}$$

Applying the same logic resulting in (21), one has the contracted explicit Euler method with update rule

$$\mathbf{c}(t_{n+1}^c) = \mathbf{c}(t_n^c) + \frac{\Delta t}{\varepsilon} \mathbf{F}(\mathbf{c}(t_n^c/\varepsilon), t_n^c/\varepsilon; \mathbf{L}, \mathcal{P}), \tag{B5}$$

where $t_n^c = t_n \varepsilon$. The only difference with (21) is the expanded time step $\Delta t^c \equiv \Delta t/\varepsilon \gg \Delta t$ which requires, as the explicit Euler method is *not* unconditionally stable, Δt sufficiently for the method to be stable.

Because the number unknowns is small in $[\mathrm{FK}]_d$ and $[\mathrm{HD}]_d$, n and $2n$ respectively, we have chosen to use (22) for a smaller step size. Conversely for $[\mathrm{SMO}]_d$, due to the large number of nN unknowns, we have opted for (B5) by choosing ε in the following manner: fix $N^c \ll N$ then

$$\varepsilon \equiv \frac{T^c}{T} = N^c \frac{\Delta t}{T}.$$

Finally, note that another alternative to time contraction is using an adaptive time step size, in which Δt is chosen at each iteration based on the estimated error from the previous step[s]. [21]

B.3 Positivity asymptotic constraints

The asymptotic values in (35) and (36) imply some constraints between the $[\mathrm{SMO}]_c$ parameters by imposing their positivity.

Consider (36), then one has

$$M_\infty = \frac{\gamma}{\mu} - m_\infty > 0 \implies m_\infty < \frac{\gamma}{\mu}, \tag{B6}$$

while (35) is more involved with

$$\begin{aligned} P_\infty = \frac{\gamma - \mu m_\infty - 2\kappa v m_\infty^2}{a m_\infty} > 0 \\ \overset{am_\infty > 0}{\Longrightarrow} \gamma - \mu m_\infty - 2\kappa v m_\infty^2 > 0 \\ \implies 2\kappa v m_\infty^2 + \mu m_\infty - \gamma < 0, \end{aligned} \tag{B7}$$

whose solution is $m_\infty^- < m_\infty < m_\infty^+$, with

$$m_\infty^\pm = \frac{-\mu \pm \sqrt{\mu^2 + 8\gamma\kappa v}}{4\kappa v}, \tag{B8}$$

which reduces to $m_\infty < m_\infty^+$ as $m_\infty^- < 0$ and $m_\infty > 0$.

Both (B6) and (B7) could be used to prove the only coherent root in (38) is the smallest real one; however, this is only possible if its analytical solutions are known meaning, due to their complexity (§ 2.3.3), this is not a viable strategy. Nonetheless these constraints help weed out potential m_∞ which is still useful.

B.4 System of Moments

The only equations that need to be proven in system (32) are (32b, 32c), since (32a) is already obvious.

In (32b) the only unclear identity is

$$f \left[\underbrace{\sum_{i,j=2}^{\infty} c_{i+j}}_{\text{(IP)}} - \frac{1}{2} \underbrace{\sum_{i=3}^{\infty} (i-3) c_i}_{\text{(IIP)}} \right] = \frac{f}{2}(M - 3P + c_2) \tag{B9}$$

where the sum goes to infinity because, as explained in § 2.3.3, assumption A1 is disregarded.

By rearranging the (I) in (B9) it holds (IP) = (IIP). Indeed, consider $n = i + j$ even, then the (i,j) asymmetric pair are $n/2 - 1$ while there is one symmetric pair $(n/2, n/2)$; remembering that i and j start at 2, the total number of pairs for $n \ge 4$ are

$$\overbrace{2(n/2 - 1 - 2 + 1)}^{\text{asym. pairs}} + \overbrace{1}^{\text{sym. pair}} = n - 3.$$

On the contrary, given $n=i+j$ odd, one has only asymmetric pairs which are unique up until $n/2$, but since $n/2$ is rational the real upper limit is $(n-1)/2$; thus, the total number of pairs are

$$\overbrace{2[(n-1)/2-2+1]}^{\text{asym. pairs}}=n-3.$$

Therefore, the total number of pairs is $n-3$ regardless the type of integer considered; this can be even visually proved using the following array:

$$\begin{array}{c|ccccc} {}_{i}\diagdown^{j} & 2 & 3 & 4 & 5 & \cdots \\ \hline 2 & 4 & 5 & 6 & 7 & \cdots \\ 3 & 5 & 6 & 7 & 8 & \cdots \\ 4 & 6 & 7 & 8 & 9 & \cdots \\ 5 & 7 & 8 & 9 & 10 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{array}. \tag{B10}$$

Hence (I) becomes

$$(\text{IP})=\sum_{n=4}^{\infty}(n-3)c_n,$$

while the sum $(\text{IP})+(\text{IIP})$ reduces to

$$\sum_{i=4}^{\infty}(i-3)c_i=\sum_{i=4}^{\infty}ic_i-3\sum_{i=4}^{\infty}c_i=M-3P+c_2, \tag{B11}$$

where c_2 and c_3 where summed and subtracted accordingly, proving (B9).

■

In (32c) the only unclear identity is

$$f\left[\underbrace{\sum_{i=2}^{\infty}i\sum_{j=2}^{\infty}c_{i+j}}_{(\text{IM})}-\frac{1}{2}\underbrace{\sum_{i=3}^{\infty}i(i-3)c_i}_{(\text{IIM})}\right]=0. \tag{B12}$$

Thanks to distributive property (IM) can be rewritten as

$$(\text{IM})=\sum_{i=2}^{\infty}\sum_{j=2}^{\infty}ic_{i+j},$$

from which it's clear the counting done in (B10) is still valid with a caveat: not all c_{i+j} with the same $i+j$ are equal, but they are weighted by i, *i.e.*:

$$(\text{IM})=\sum_{n=4}^{\infty}\sum_{i=2}^{n-2}ic_n=\sum_{n=4}^{\infty}c_n\sum_{i=2}^{n-2}i=\sum_{n=4}^{\infty}c_n\left(\sum_{i=1}^{n-2}i-1\right),$$

where the upper limit of the second sum w.r.t. i comes from the fact that each pair has to be summed starting with $(i,j)=(2,n-2)$ and ending with $(i,j)=(n-2,2)$; a visual representation of what has been said can be found in the following array:

$$\begin{array}{c|ccccc} {}_{j}\diagdown^{n} & 4 & 5 & 6 & 7 & \cdots \\ \hline 2 & 2 & 3 & 4 & 5 & \cdots \\ 3 & & 2 & 3 & 4 & \cdots \\ 4 & & & 2 & 3 & \cdots \\ 5 & & & & 2 & \cdots \\ \vdots & & & & & \ddots \end{array}. \tag{B13}$$

By remembering the identity $\sum_{i=1}^{n}i=[(n+1)n]/2$, (IM) becomes

$$\begin{aligned}(\text{IM})&=\sum_{n=4}^{\infty}c_n\left(\frac{(n-1)(n-2)}{2}-1\right)=\sum_{n=4}^{\infty}c_n\left(\frac{n^2-3n+2}{2}-1\right)\\&=\frac{1}{2}\sum_{n=4}^{\infty}n(n-3)c_n=\frac{1}{2}\sum_{n=3}^{\infty}n(n-3)c_n=(\text{IIM}),\end{aligned}$$

proving (B12).

■

B.5 Asymptotic size distribution

Assume to know the asymptotic values of P_∞ and m_∞ in (35, 38) and modify the equations $i>2$ in $[\text{SMO}]_c$ to

$$\frac{\mathrm{d}c_i}{\mathrm{d}t}=-\mu c_i+ac_1(c_{i-1}-c_i)-\frac{(i-3)}{2}fc_i+f\sum_{j=1}^{\infty}c_{i+j}, \tag{B14}$$

where

- ◇ the summation starts at 1 and not 2, *i.e.* A7 is modified to allow fragmentation of polymers into monomers,
- ◇ A1 is ignored to have $L\to\infty$ and
- ◇ the system is homogeneous in space.

By subtracting the $(i+1)$-th and i-th (B14) equations, one finds

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}(c_{i+1}-c_i)=&-\left[\mu+\frac{f}{2}(i-2)\right](c_{i+1}-c_i)\\&+am(c_{i+1}-2c_i+c_{i-1})\\&-fc_{i+1}-\frac{f}{2}c_i,\end{aligned} \tag{B15}$$

with $c_1 \equiv m$, as done in (32). Define

$$c(s,t) : \mathbb{R}^+ \times \mathbb{R}^+ \to \mathbb{R}^+$$

such that $c(i,t) = c_i(t)$ for $i \in \mathbb{N}^+$, then using its discretization with a unit step, that is

$$\frac{\partial c}{\partial s} \approx c_{i+1} - c_i,\ \frac{\partial^2 c}{\partial s^2} \approx c_{i+1} - 2c_i + c_{i-1},\ c_{i+1} \approx \frac{\partial c}{\partial s} + c,$$

and[II] $c \approx c_i$, (B15) becomes

$$\frac{\partial^2 c}{\partial s \partial t} = -\left(\mu + \frac{f}{2}s\right)\frac{\partial c}{\partial s} - am\frac{\partial^2 c}{\partial s^2} - 3\frac{f}{2}c, \tag{B16}$$

whose steady state form is given by

$$am_\infty \frac{\partial^2 c}{\partial s^2} + \left(\mu + \frac{f}{2}s\right)\frac{\partial c}{\partial s} + 3\frac{f}{2}c = 0. \tag{B17}$$

Setting now

$$\nu_1 = \frac{\mu}{am_\infty} \quad \text{and} \quad \nu_2 = \frac{1}{am_\infty}\frac{f}{2}, \tag{B18}$$

and changing variable [12, (B13) on p. 3473] from s to

$$u = \frac{\nu_1 + \nu_2 s}{\sqrt{\nu_2}}, \tag{B19}$$

one can transform (B17) into

$$\frac{\partial^2 c}{\partial u^2} + u\frac{\partial c}{\partial u} + 3c = 0. \tag{B20}$$

Assume now c has form

$$c^\infty(u) = w(u)\exp\left[-\frac{u^2}{4}\right], \tag{B21}$$

for some function $w(u)$, then substituting this ansatz in (B20) leads to [12, (B15) on p. 3473]

$$\frac{\partial^2 w}{\partial u^2} + \left(2 + \frac{1}{2} - \frac{u^2}{4}\right)w = 0, \tag{B22}$$

namely the parabolic cylinder equation with index $\nu = 2$ [10, (3.5.11) at p. 96] with solution

$$w(u) = N\mathrm{He}_2(u)\exp\left[-\frac{u^2}{4}\right], \tag{B23}$$

where N is a constant and $\mathrm{He}_2(u) = u^2 - 1$ is the second Hermite polynomial [10, p. 99]. Expanding (B18, B19, B23) into (B21) gives [15, (76) on p. 11]

$$\begin{aligned} c^\infty(s) &= N\left[\frac{(2\mu + sf)^2 - 2afm_\infty}{2afm_\infty}\right]\exp\left[-\frac{(2\mu+sf)^2}{4afm_\infty}\right], \\ &\equiv K\left[(2\mu+sf)^2 - 2afm_\infty\right]\exp\left[-\frac{(2\mu+sf)^2}{4afm_\infty}\right], \end{aligned}$$

[II] This last approximation is not contradictory because the discretization considers c_i as the reference point: the previous first two approximations are its forward and central difference formulas, while the last one is the Taylor expansion centered in $s = i$ at a fixed t, confirming $c \approx c_i$.

where $K \equiv N/(2afm_\infty)$ and can be found by considering the continuous version of

$$P_\infty = \sum_{l=2}^{\infty} c_l^\infty \implies P_\infty = \int_2^\infty c^\infty \, \mathrm{d}s. \tag{B24}$$

Before integrating it's useful to change integration variable to $z = 2\mu + sf$ with

$$\int_2^\infty c^\infty \, \mathrm{d}s \implies \frac{1}{f}\int_{2\mu+2f}^\infty c^\infty \, \mathrm{d}z,$$

then

$$P_\infty = \frac{K}{f}\Bigg(\overbrace{\int_{2\mu+2f}^\infty z^2 \exp\left[-\frac{z^2}{4afm_\infty}\right]\mathrm{d}z}^{\text{(I)}} \underbrace{-2afm_\infty\int_{2\mu+2f}^\infty \exp\left[-\frac{z^2}{4afm_\infty}\right]}_{\text{(II)}}\Bigg),$$

where (I) can be integrated by parts

$$\begin{aligned} \text{(I)} = &-2afm_\infty\left[z\exp\left(-\frac{z^2}{4afm_\infty}\right)\right]_{2\mu+2f}^\infty \\ &+2afm_\infty\int_{2\mu+2f}^\infty \exp\left[-\frac{z^2}{4afm_\infty}\right], \end{aligned}$$

whose last term cancels out with (II), leaving

$$\begin{aligned} P_\infty &= \frac{K2a\cancel{f}m_\infty}{\cancel{f}}\left[z\exp\left(-\frac{z^2}{4afm_\infty}\right)\right]_\infty^{2\mu+2f} \\ &= K4am_\infty(\mu+f)\exp\left[-\frac{(\mu+f)^2}{afm_\infty}\right], \end{aligned}$$

which gives [15, (77) on p. 11]

$$K = \frac{P_\infty \exp\left[\frac{(f+\mu)^2}{afm_\infty}\right]}{4am_\infty(f+\mu)}. \tag{B25}$$

By using (B23, B25) in (B21), the asymptotic distribution (41) is derived.

■

Finally, (42) is obtained simply by finding the critical point of (B21) through derivation w.r.t. s.